Declining Business Dynamism in Korea: Shocks, Responsiveness, and the Role of Mature Firms*

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Abstract

Business dynamism has declined sharply in Korea: job reallocation, entry, and exit have all fallen by roughly half since the early 2000s. Following Decker et al. (2020), we ask whether businesses face smaller idiosyncratic shocks or respond less to them. Productivity dispersion did not systematically decline, innovation dispersion fell only modestly, and measured persistence rose, offering at most limited support for a shock-based explanation. Because the standard growth measure embeds exit, we separate employment adjustment from survival. Employment adjustment weakens most among young establishments, as in the United States; survival selection, by contrast, deteriorates among mature establishments, a pattern that survives a fixed-cohort test. Restoring both margins to their 2000s responsiveness recovers about three-fifths of the reallocation decline and nearly one percentage point of annual labor productivity growth.

Keywords: business dynamism; job reallocation; responsiveness; productivity shocks; firm age; creative destruction

JEL codes: D24, E24, L25, O47

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1 Introduction

Reallocation from shrinking, low-productivity businesses toward expanding, high-productivity ones is a central source of aggregate productivity growth (Schumpeter, 1942; Foster et al., 2001). Yet business dynamism—job reallocation, entry, and exit—has declined across advanced economies (Decker et al., 2014, 2020; Bijnens and Konings, 2020; Biondi et al., 2025). Korea is an unusually stark case. Between 2001 and 2023, the private-sector job reallocation rate fell from roughly 68 percent to about 40 percent, while entry and exit rates roughly halved. The decline predates the pandemic and coincides with Korea's broader growth slowdown: potential growth fell from around 5 percent in the early 2000s to about 2 percent for 2024–2026 (Bank of Korea, 2024), while manufacturing TFP growth slowed sharply (Lee et al., 2020).

We ask why Korean dynamism declined, which businesses drove the change, and whether the resulting micro changes matter for aggregate outcomes. We apply the framework of Decker et al. (2020) (henceforth DHJM), which distinguishes a *shock* explanation—smaller or less persistent idiosyncratic productivity shocks—from a *responsiveness* explanation—a flatter mapping from productivity to growth and exit. Our establishment panel combines the universe *Census on Establishments* with manufacturing production data and auxiliary firm-level data for manufacturing and services; detailed data construction is reported in Appendix A.

Our first result is that the available evidence provides at most limited support for a simple shock-based explanation. Productivity dispersion did not systematically decline after 2010, including within narrowly defined industries, and measured persistence increased. Estimated innovation dispersion declines modestly, so these moments are not a sharp rejection of all shock-based mechanisms. Responsiveness, however, weakens clearly: employment growth becomes less sensitive to productivity and the productivity–exit gradient flattens. These patterns survive the exclusion of the global financial crisis and the pandemic and extend beyond manufacturing establishments to firm-level data and services.

Interpreting the coefficients requires separating two measurement issues. The DHS growth rate assigns exiters a growth rate of -2, so measured growth responsiveness mechanically contains the exit margin. On continuing establishments, employment adjustment still weakens, but by less. In addition, linear exit slopes scale with the exit rate itself, which more than halved over the sample. Scale-adjusted exit estimates therefore serve as robustness checks on magnitude rather than as the main specification. Across methods, the pooled behavioural share of the raw exit-gradient decline is roughly one-seventh to one-quarter, while the age-specific result is considerably sharper.

That age pattern is the paper's main result. On the employment-adjustment margin, Korea resembles the United States: both age groups become less responsive, with the larger decline among young establishments. On the survival margin, however, the weakening is concentrated among mature establishments. Mature businesses selected more sharply on productivity in the 2000s, but their productivity–survival link weakened substantially thereafter, while the corresponding relationship among young establishments is statistically unchanged. The scale-free probit estimate implies about a 41 percent decline for mature establishments and essentially no change for young ones. A fixed-cohort test shows that this is not a composition effect. The evidence also weakens, but does not eliminate, a pure vintage interpretation. A shift-share decomposition reaches the same conclusion from the aggregate side: changes in the age distribution account for less than one-third of the fall in reallocation.

For shorthand, we refer to the weakening of productivity-based exit selection as the *survival wedge*. The term describes the flattening of the productivity–survival relationship; it is not a separately estimated establishment-level wedge. Its age profile narrows the set of plausible mechanisms. A mature-specific employment-adjustment friction fits poorly because the continuers-only employment decline is stronger among young establishments. Age-neutral changes in competitive toughness also do not naturally generate the survival-side age asymmetry, and cross-industry evidence on concentration and exposure to Chinese competition provides little support for that explanation. Possible alternatives include policy finance, credit guarantees, relationship lending, or other forms of forbearance whose incidence rises with operating history. We treat these as candidate mechanisms, not identified causes.

The aggregate implications are economically meaningful. Restoring both the employment-adjustment and survival margins to their 2000s responsiveness recovers about three-fifths of the decline in manufacturing job reallocation and nearly one percentage point of annual labor-productivity growth; the survival margin contributes about half of the reallocation effect.

The paper makes three contributions. First, it applies the DHJM shocks-versus-responsiveness framework to long-run Korean microdata and shows that declining responsiveness is central to the Korean episode, while the shock evidence is mixed. Second, it documents a new age asymmetry: young establishments lose more employment-adjustment responsiveness, whereas the distinctive weakening in productivity-based survival selection is concentrated among mature incumbents. Third, it shows why the standard growth and exit measures require separate interpretation and connects the resulting micro estimates to aggregate reallocation and productivity outcomes.

Our baseline productivity measure is revenue-based TFP, which combines technical efficiency with idiosyncratic demand conditions (Foster et al., 2008). This is the relevant

profitability signal for the responsiveness question, and we also report robustness to control-function TFP and labor productivity. The paper identifies where responsiveness weakened, not the structural primitive behind it.

The remainder of the paper proceeds as follows. Section 2 relates the paper to the literature. Section 3 defines the main measures, with data sources and sample construction in Appendix A. Section 4 documents the decline in dynamism. Section 5 tests the shock and responsiveness hypotheses. Section 6 studies heterogeneity by business age. Section 7 discusses candidate mechanisms. Section 8 quantifies the aggregate implications, and Section 9 concludes.

2 Related Literature

Our paper builds on three strands of research. The first is the classic literature on job flows and reallocation-driven productivity growth. Davis et al. (1996) used establishment-level data for U.S. manufacturing to document the magnitudes of simultaneous job creation and destruction and established the measurement conventions we adopt. Foster et al. (2001) showed that the reallocation of activity across establishments—including the exit of low-productivity businesses and the entry of new ones—accounts for a substantial share of aggregate productivity growth, giving empirical content to the Schumpeterian notion of creative destruction (Schumpeter, 1942). Hopenhayn and Rogerson (1993) provided the canonical general-equilibrium framework in which policy-induced adjustment frictions, such as firing taxes, depress reallocation and aggregate productivity.

The second strand documents and interprets the decline in business dynamism. Decker et al. (2014) established the U.S. facts on declining entrepreneurship and reallocation. Decker et al. (2020) then developed the identification strategy we apply: in a broad class of firm-dynamics models, the pace of reallocation depends on the dispersion and persistence of idiosyncratic productivity shocks and on the marginal responsiveness of businesses to those shocks; changes in dynamism can therefore be attributed to "shocks" or to "responsiveness" by estimating both objects in microdata. For the United States they find rising within-industry dispersion alongside falling responsiveness—concentrated among young businesses in high-tech sectors—and interpret the pattern as rising adjustment frictions. Akcigit and Ates (2021) rationalize declining U.S. dynamism in an endogenous growth model with declining knowledge diffusion from leaders to followers, and Klenow and Li (2024) document that entry costs rise systematically with development, connecting slower entry to growth itself. For Europe, Biondi et al. (2025) apply the DHJM framework to micro-aggregated data

for nineteen countries and find that declining responsiveness *and* shrinking shock dispersion both contribute, with the reallocation decline concentrated among large and mature firms; Bijnens and Konings (2020) document parallel trends for Belgium. Relative to this work, Korea turns out to be a particularly clean case: shock dispersion did not fall, so the decline loads entirely on responsiveness, and we push the heterogeneity analysis one step further by estimating age-specific responsiveness directly.

The third strand concerns firm demographics and Korean evidence. Haltiwanger et al. (2013) showed for the United States that, conditional on age, small businesses do not systematically outgrow large ones—growth is a young-firm phenomenon—and Cho et al. (2017) confirmed the same pattern in Korean data. For Korea, Kim (2017) attributed a substantial part of the manufacturing growth slowdown to weaker productivity growth among young establishments, and Woo (2020) documented stagnating value-added growth across establishment age groups. Kim and Son (2015) found that smaller and younger firms exhibit stronger job creation, while Cho (2017) linked declining allocative efficiency to business-group-affiliated firms. Lee and Won (2021) showed that the speed of job reallocation in Korean manufacturing slowed after the Asian financial crisis and that the exit of low-productivity establishments diminished, and Kim and Hong (2022) connected the aggregate productivity slowdown to deteriorating reallocation among incumbents. Lee and Shin (2024) take a long-run view of Korean plant dynamics over six decades and document a broad decline in dynamism accompanying the growth slowdown.

Our contribution relative to the Korean literature is twofold. First, whereas existing studies primarily measure dynamism or estimate its consequences for productivity, we identify *why* dynamism declined, implementing the DHJM decomposition between shocks and responsiveness on Korean microdata. Second, we provide the first direct evidence on the age profile of the responsiveness decline. The result—that the distinctive weakening of productivity-based survival selection is concentrated among mature incumbents—qualifies the presumption inherited from the U.S. literature and from Korean studies emphasizing young firms, and redirects attention toward the restructuring capacity of incumbents.

3 Data and Measurement

The empirical analysis combines universe establishment records with manufacturing production data and an auxiliary firm-level panel. Appendix A describes the underlying sources, sample construction, coverage restrictions, panel linkage, and variable definitions in detail. We focus here on the measurement of business dynamism and productivity.

3.1 Measuring business dynamism

Following Davis et al. (1996) and Foster et al. (2001), we define business dynamism as the job reallocation rate: the sum of job creation and job destruction as a share of employment. Let E_{it} denote employment of establishment i in year t, and define the symmetric growth rate

$$g_{it} \;=\; \frac{E_{i,t+1} - E_{it}}{\frac{1}{2}\left(E_{i,t+1} + E_{it}\right)}, \tag{1}$$

which lies in $[-2, 2]$ and accommodates entry ($g_{it} = 2$) and exit ($g_{it} = -2$). With $X_{st} = \sum_{i \in s} \frac{1}{2}(E_{i,t+1} + E_{it})$ denoting size-weighted employment in industry s, the job creation and destruction rates are

$$JC_{st} \;=\; \sum_{i \in s,\; g_{it} > 0} \frac{E_{i,t+1} - E_{it}}{X_{st}}, \qquad JD_{st} \;=\; \sum_{i \in s,\; g_{it} < 0} \frac{|E_{i,t+1} - E_{it}|}{X_{st}}, \tag{2}$$

and the job reallocation rate is

$$JR_{st} \;=\; JC_{st} + JD_{st}. \tag{3}$$

As complementary count-based measures, the entry rate in year t is the number of newly appearing establishments divided by the number of active establishments, and the exit rate is the number of establishments present in $t-1$ but absent in t divided by the number of active establishments.[1]

3.2 Measuring establishment productivity

Following DHJM, our baseline productivity measure is a revenue-based establishment TFP that jointly reflects technical efficiency and idiosyncratic demand conditions. Consider an establishment facing demand $R_{it} = D_{it} Q_{it}^{\zeta}$, where D_{it} is an idiosyncratic demand shifter and output is $Q_{it} = A_{it} F(X_{it})$, with A_{it} technical efficiency and X_{it} a vector of inputs. Log revenue can then be written as

$$\ln R_{it} \;=\; \sum_{x} \alpha_x \ln X_{x,it} \;+\; z_{it}, \tag{4}$$

where α_x is the revenue elasticity of input x. Under static profit maximization, α_x equals the share of expenditure on input x in total revenue, so the productivity residual—our baseline

[1] Because the Census on Establishments is a complete count, exit in year t is defined as presence in $t-1$ and absence in t. The 2020 wave expanded the census population frame; to limit the resulting selection effect, the 2020 entry rate is replaced by interpolation.

Table 1: Summary statistics for manufacturing establishment productivity

	TFPS	TFPP	ln(LP)
Mean	2.90	2.96	5.39
Standard deviation	0.53	0.59	0.77

Notes: Authors' calculations from the Mining and Manufacturing Survey and Economic Census microdata (Statistics Korea). Each measure is trimmed at the 0.1st and 99.9th percentiles. TFPS is revenue TFP computed from equation (5) with factor revenue elasticities measured by expenditure shares; TFPP is estimated with the control-function method of Ackerberg et al. (2015); LP is real value-added per worker.

measure, denoted TFPS—is

$$z_{it} \equiv \ln \mathrm{TFPS}_{it} = \ln R_{it} - \sum_{x} \alpha_x \ln X_{x,it}. \tag{5}$$

Because z_{it} is revenue-based, it conflates technical efficiency with demand and price variation (Foster et al., 2008). For our question this composite is a feature rather than a bug: a profit-maximizing business adjusts employment in response to profitability, whether the underlying impulse is a technology improvement or a demand shift, so revenue TFP is precisely the signal to which responsiveness should be measured (Decker et al., 2020). We nonetheless assess robustness in two ways. First, we estimate production-function-based TFP (denoted TFPP) using the control-function approach of Levinsohn and Petrin (2003) as refined by Ackerberg et al. (2015), which does not rely on static first-order conditions, though it demands long panels and many establishments per industry for precise elasticity estimates. Second, we use log real value-added per worker (labor productivity, LP) as a transparent auxiliary measure.[2] To limit the influence of outliers, each productivity measure is trimmed at the 0.1st and 99.9th percentiles.

Table 1 reports summary statistics. The two TFP measures behave similarly, and their dispersion is large: a standard deviation of 0.53 implies that an establishment one standard deviation above its industry mean is roughly $e^{0.53} \approx 1.7$ times as revenue-productive as the average establishment—dispersion of the magnitude familiar from the misallocation literature (Hsieh and Klenow, 2009). Labor productivity is more dispersed still, reflecting the omission of capital and intermediate inputs.

[2]Labor productivity is not an exogenous shock measure—employment appears in its denominator—but under adjustment costs it co-moves strongly and positively with fundamental profitability shocks, making it a useful proxy (Decker et al., 2020).

4 The Decline of Business Dynamism in Korea

4.1 Job reallocation

Figure 1 plots the job reallocation rate for 2001–2023. Dynamism has declined steadily and substantially: the economywide reallocation rate fell from roughly 68 percent in 2001 to about 40 percent in 2023, a drop of 28 percentage points. The decline appears in both sectors—about 21 percentage points in manufacturing and about 38 percentage points in services. Crucially, the trend is not an artifact of the pandemic: comparing 2001 with 2019, the declines are already 28 percentage points economywide, 23 in manufacturing, and 36 in services. Nor is it visible only in reallocation: administrative business demography statistics show the business turnover rate falling from about 30 percent in 2007 to below 25 percent in recent years. The persistence of the decline through normal times, crisis times, and recovery times points to structural rather than cyclical forces.

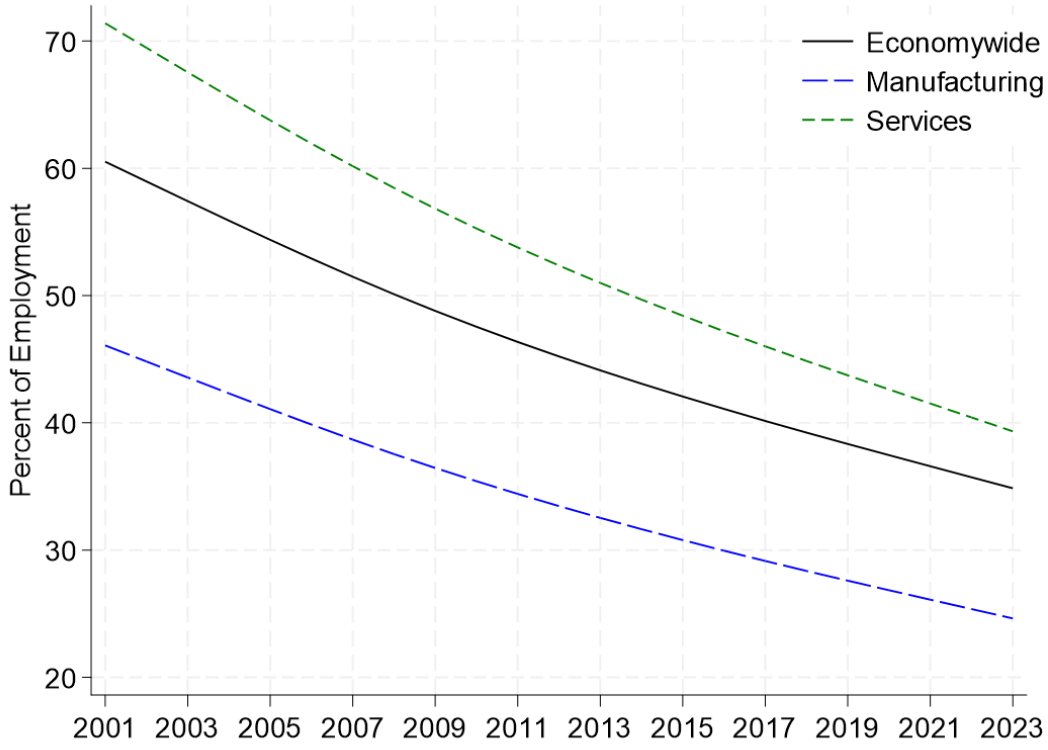


Figure 1: Business dynamism in Korea: the job reallocation rate, 2001–2023

Notes: Authors' calculations from the Census on Establishments microdata (2000–2023). "All industries" comprises manufacturing and services. The job reallocation rate is defined in equations (1)–(3). Series shown are HP trends with smoothing parameter 6.25.

4.2 Entry and exit

Figure 2 shows that the count-based margins mirror the reallocation series. In services, the entry rate fell from about 43 percent in 2002 to about 24 percent in 2023, and the exit rate from 46 to 18 percent. In manufacturing, the entry rate fell from 29 to 15 percent and the exit rate from 32 to 16 percent. Both rates remain higher in services than in manufacturing, but the declines are larger in services. Taken together, Figures 1 and 2 document a broad-based, two-decade erosion of the entry, growth, contraction, and exit processes through which resources are reallocated across Korean businesses.

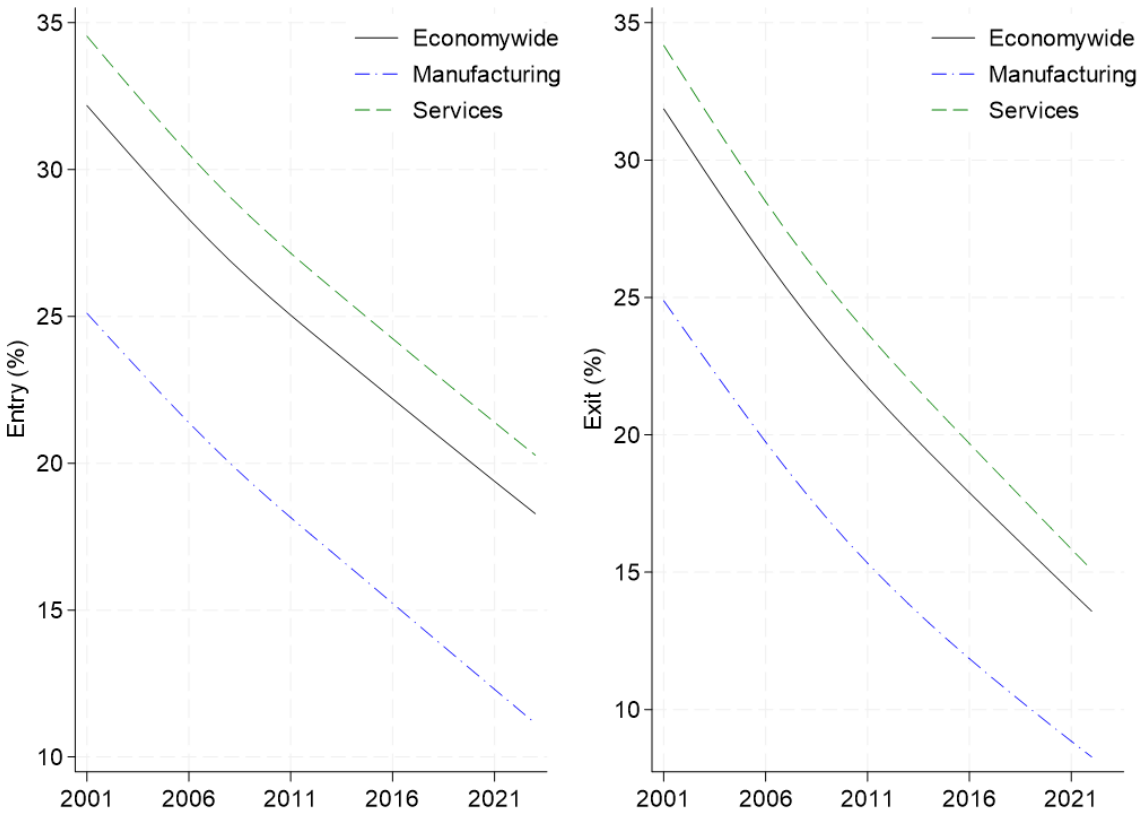


Figure 2: Establishment entry and exit rates, 2001–2023

Notes: Authors' calculations from the Census on Establishments microdata (2001–2023). The entry (exit) rate is the number of entering (exiting) establishments divided by the number of active establishments. Series shown are HP trends with smoothing parameter 6.25. Because the 2020 wave expanded the census population frame, the 2020 entry rate is replaced by interpolation to limit selection effects.

5 Shocks versus Responsiveness

5.1 Conceptual framework

Why might reallocation decline? DHJM show that, in a broad class of firm-dynamics models in the tradition of Hopenhayn and Rogerson (1993), the pace of within-industry reallocation is governed by two objects: the distribution of idiosyncratic productivity shocks that businesses face, and the marginal responsiveness of businesses to those shocks. A simple static example fixes ideas. Suppose an establishment's revenue function is $R_{it} = Z_{it} L_{it}^{\phi}$ with $\phi \in (0,1)$, where Z_{it} combines technical efficiency and demand conditions and the curvature ϕ reflects decreasing returns or imperfect competition. Absent adjustment frictions, the first-order condition for labor implies

$$\ln L_{it}^{*} \;=\; \frac{1}{1-\phi}\,\ln\!\left(\frac{\phi Z_{it}}{W_t}\right), \tag{6}$$

so that employment growth is increasing in current productivity and decreasing in initial size:

$$g_{it} \;\approx\; \kappa_t \;+\; \frac{1}{1-\phi}\,z_{it} \;-\; \ell_{i,t-1}, \tag{7}$$

where $z_{it} = \ln Z_{it}$ and $\ell_{i,t-1} = \ln L_{i,t-1}$. More generally, with adjustment costs and market power the policy function can be written $g_{it} = f_t(z_{it}, \ell_{it-1})$, where the mapping f_t may vary over time.

This formulation nests the two hypotheses. Under the *shock* hypothesis, reallocation falls because the dispersion of z_{it} shrinks or its persistence weakens, while f_t is stable. Under the *responsiveness* hypothesis, the distribution of shocks is stable but f_t flattens—as would occur with rising adjustment costs, tighter labor market frictions, growing market power, or intensified tax-and-subsidy distortions—so that any given productivity signal elicits less employment adjustment and less selection on exit.

5.2 Testing the shock hypothesis

We first ask whether the shock process changed. We split the sample at 2010, the point at which the aggregate productivity slowdown set in and the decline in dynamism was steepest, and compare the dispersion and persistence of establishment productivity across subperiods.[3] Persistence is estimated from the AR(1) process $z_{it} = \rho_z z_{i,t-1} + \varepsilon_{it}$, period by period, for

[3] Concerns that the 2020–2023 years conflate the pandemic are addressed by computing the same statistics with the COVID period separated; the changes in dispersion and AR(1) persistence are similar to those shown in Figure 3.

both raw and within-industry productivity.[4]

The available evidence provides at most limited support for the shock hypothesis. A simple shock-based account would predict a systematic compression in productivity dispersion or lower persistence. Figure 3 (Panel A) shows instead that the raw standard deviation of manufacturing TFPS rose from 0.498 before 2010 to 0.537 after, while within-industry dispersion changed little (0.351 to 0.338 at the two-digit level and 0.342 to 0.333 at the three-digit level); within-industry labor-productivity dispersion rose modestly. Panel B shows that measured persistence ρ_z also increased slightly after 2010.[5] These moments therefore do not support a simple account in which a substantially quieter idiosyncratic environment is the main driver of declining reallocation, although they do not rule out a role for changes in the innovation process. This contrasts with Europe, where Biondi et al. (2025) find that declining shock dispersion and declining responsiveness both contribute.

5.3 Responsiveness in Employment and Exit

We now estimate responsiveness directly, using the log-linear approximation of the policy function γ_t:

$$g_{it} \;=\; \beta_t\, z_{it} \;+\; \lambda_t\, \ell_{it} \;+\; \Gamma' X_{it} \;+\; \varepsilon_{it}, \tag{8}$$

where g_{it} is the DHS employment growth rate between t and $t+1$ from equation (1), z_{it} is establishment TFPS realized at the start of year t, and ℓ_{it} is log employment at the start of year t. The control vector X_{it} includes industry effects, year effects, industry-by-year effects, establishment size and age controls, and the interaction of employment with productivity. The responsiveness coefficient β_t is allowed to vary over time in four ways:

$$\beta_t \in \begin{cases} \beta_0 + \beta_\tau\, t & \text{(linear trend)}, \\ \beta_1 + \beta_2\, \mathbf{1}\{t \geq 2010\} & \text{(post-2010 shift)}, \\ \beta_1 \mathbf{1}\{t \in 2000\text{s}\} + \beta_2 \mathbf{1}\{t \in 2010\text{s}\} + \beta_3 \mathbf{1}\{t \in 2020\text{s}\} & \text{(by decade)}, \\ \text{by decade, excluding crisis years} & \text{(“no-crisis”)}, \end{cases} \tag{9}$$

[4] Within-industry productivity is defined as $\widetilde{z}_{it} = z_{it} - \overline{z}_{s(i)t}$, deviations from industry-year means at the two- and three-digit levels of the Korean Standard Industrial Classification.

[5] As an additional check, the estimated standard deviation of the AR(1) innovation declines modestly after 2010: from 0.219 to 0.203 for TFPS after removing two-digit industry-year means, with a similar pattern at the three-digit level, and from 0.403 to 0.365 for TFPP. We therefore view the shock evidence as mixed rather than as a sharp rejection; the decline in innovation dispersion is modest relative to the pronounced fall in measured responsiveness.

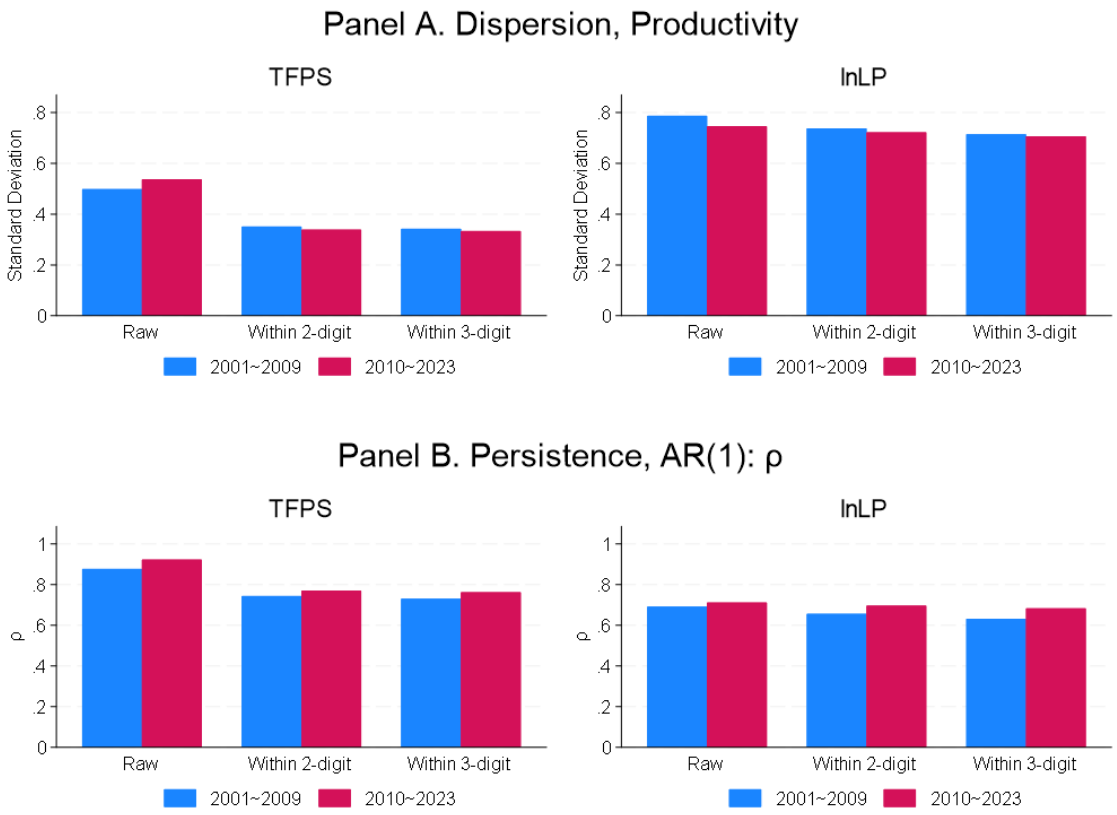


Figure 3: Dispersion and persistence of establishment productivity, before and after 2010

Notes: Authors' calculations from the Mining and Manufacturing Survey and Economic Census microdata. "Within" panels use deviations of TFPS and lnLP from two- and three-digit industry means. Persistence ρ_z is estimated from $z_{it} = \rho_z z_{i,t-1} + \varepsilon_{it}$ by subperiod.

where the no-crisis sample excludes 2008–2010 and the pandemic period (2020 onward), retaining 2001–2007 and 2011–2019. λ_t is allowed to vary according to the corresponding specification for β_t[6].

Table 2 reports the estimates for manufacturing establishments. At the start of the sample, responsiveness is strong: the baseline coefficient is 0.122, so more productive establishments systematically grow faster—the virtuous selection mechanism at the heart of creative destruction. But responsiveness declines steadily. The linear-trend specification (column 1) implies a loss of 0.002 per year; the post-2010 shift (column 2) is −0.023 and highly significant; and the decade specification (column 3) shows a stepwise fall from 0.114 in the 2000s to 0.094 in the 2010s and 0.080 in the 2020s, with all pairwise differences statistically significant. Column 4 shows that the 2000s-to-2010s decline survives, essentially unchanged, when crisis years are excluded, confirming that it is a structural trend rather than a crisis artifact.[7]

The level of responsiveness in Korea is notably below the U.S. benchmark: DHJM report a coefficient of 0.29 for U.S. manufacturing establishments before 1997, more than twice our 2000s estimate, plausibly reflecting Korea's more rigid labor market.[8]

The exit margin. The DHS growth rate already embeds exit as $g_{it} = -2$, but exit warrants its own estimate: the selective exit of low-productivity businesses is the extensive margin of responsiveness. We therefore estimate the exit analogue of equation (8), replacing the dependent variable with an indicator equal to one if establishment i, present in year t, has exited by $t + 1$, and retaining the same right-hand side.

Table 3 reports the estimates, and read at face value they are dramatic. The exit–productivity gradient falls by roughly half between the 2000s and the 2010s, and by the 2020s it is statistically indistinguishable from zero. Taken literally, these numbers say that the cleansing component of creative destruction has largely ceased to operate in Korean manufacturing.

That reading would be a mistake, and the reason matters enough that we state it here and carry it through the paper. The slope of a linear probability model is not invariant to the level of the outcome: it scales with the density of the latent index at the exit threshold, which

[6]The naming and timing conventions in (9) represent an empirical analog to equation (7) given the timing of the measurement of growth and productivity in the data.

[7]In economic terms, an establishment one standard deviation (0.53) above mean productivity grew about 6.0 percentage points faster than the average establishment in the 2000s, about 5.0 points faster in the 2010s, and about 4.2 points faster in the 2020s.

[8]The comparison is indicative rather than exact: the two studies share the establishment-level design and a very similar specification, but differ in sample period, industry classification, and the treatment of regional variation.

Table 2: Employment Responsiveness to TFPS

	(1)	(2)	(3)	(4)
z_{it}	0.122***	0.114***		
	(0.0066)	(0.0056)		
$z_{it}\times$ trend	−0.002***			
	(0.0005)			
$z_{it}\times$ Post-2010		−0.023***		
		(0.0065)		
$z_{it}\times$ 2000s: β_1			0.114***	0.120***
			(0.0056)	(0.0065)
$z_{it}\times$ 2010s: β_2			0.094***	0.097***
			(0.0040)	(0.0043)
$z_{it}\times$ 2020s: β_3			0.080***	
			(0.0051)	
p-value: $\beta_1 = \beta_2$			0.004	0.002
p-value: $\beta_1 = \beta_3$			0.000	
p-value: $\beta_2 = \beta_3$			0.027	
Observations	644,880	644,880	644,880	455,628
Sample	2001–2023	2001–2023	2001–2023	No-crisis

Notes: Authors' estimates from the Mining and Manufacturing Survey and Economic Census microdata. The dependent variable is the DHS employment growth rate between t and $t+1$; z_{it} is establishment TFPS at the start of year t. All specifications follow equation (8) and include industry effects, year effects, industry-by-year effects, establishment size and age controls, initial employment, and the employment–productivity interaction. The no-crisis sample excludes 2008–2010 and 2020 onward (retaining 2001–2007 and 2011–2019). Standard errors clustered at the establishment level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

is a pure function of the exit rate. Korea's exit rate more than halved over our sample. A large part of the fall in the estimated gradient would therefore have occurred with no change in behaviour at all—and the same arithmetic applies to any comparison of linear exit slopes across periods, countries, or groups whose exit rates differ. Appendix B.2 sets out the scaling formally and puts the two periods on a common footing in three ways: an analytic decomposition of the observed change into a base-rate component and a behavioural one; a probit, whose index coefficient lies on the latent scale directly and is therefore numerically commensurable with that decomposition; and a non-parametric relative-hazard profile that invokes no index assumption at all. The three agree. Exit selectivity did weaken, but by appreciably less than the raw slopes suggest. We report the raw linear estimates throughout for comparability with DHJM and with the European literature, and we work with the scale-free selectivity parameter, denoted θ, wherever a magnitude is being interpreted.

The two margins are also not independent of one another, since the growth measure contains exit by construction. We defer that decomposition to Section 6.3, because it interacts with establishment age and is best presented alongside the age-specific estimates.

Table 3: Exit Responsiveness to TFPS

	(1)	(2)	(3)	(4)
z_{it}	−0.020***	−0.017***		
	(0.0029)	(0.0025)		
$z_{it}\times$ trend	0.001***			
	(0.0002)			
$z_{it}\times$ Post-2010		0.011***		
		(0.0028)		
$z_{it}\times$ 2000s: β_1			−0.017***	−0.019***
			(0.0025)	(0.0029)
$z_{it}\times$ 2010s: β_2			−0.007***	−0.009***
			(0.0016)	(0.0017)
$z_{it}\times$ 2020s: β_3			−0.004	
			(0.0019)	
p-value: $\beta_1 = \beta_2$			0.001	0.003
p-value: $\beta_1 = \beta_3$			0.000	
p-value: $\beta_2 = \beta_3$			0.148	
Observations	672,366	672,366	672,366	472,640
Sample	2001–2023	2001–2023	2001–2023	No-crisis

Notes: Authors' estimates from the Mining and Manufacturing Survey and Economic Census microdata. The dependent variable equals one if the establishment, present in year t, has exited by $t+1$ (linear probability model). All specifications include the full control set of equation (8). The no-crisis sample excludes 2008–2010 and 2020 onward. Standard errors clustered at the establishment level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Taken together, the pooled estimates reproduce the DHJM finding on both margins: the responsiveness of employment growth and of exit to productivity declined, in Korea as in the United States, and neither decline is an artifact of the crisis years. What the pooled estimates cannot tell us is which businesses stopped responding. That is the subject of the next section, and it is where the Korean case departs from the U.S. one.

6 Firm Age and the Decline in Responsiveness

6.1 The aging business sector and composition effects

The Korean business sector has aged rapidly. Figure 4 plots the employment share of young establishments—those aged zero to six years—which has fallen by more than ten percentage points since the mid-2000s, with a particularly sharp drop in the early 2010s. This fact suggests a natural "composition" account of declining dynamism: young businesses are the locus of up-or-out dynamics—low-productivity entrants shrink or exit quickly while high-productivity entrants grow fast—so a smaller young-business share could mechanically depress measured reallocation even if behavior within each age group were unchanged.

We evaluate the composition account with a shift-share decomposition of the change in the job reallocation rate. Writing the aggregate rate as an employment-share-weighted average across age groups, $JR_t = \sum_a \omega_{a,t} JR_{a,t}$, the change between a base period t_0 and an end period t_1 decomposes as

$$JR_{t_1} - JR_{t_0} = \underbrace{\sum_a \omega_{a,t_0}\, \Delta JR_a}_{\text{within}} + \underbrace{\sum_a JR_{a,t_0}\, \Delta\omega_a}_{\text{between}} + \underbrace{\sum_a \Delta\omega_a\, \Delta JR_a}_{\text{cross}}. \tag{10}$$

To limit cyclical influences we compare the averages of 2005–2007 and 2015–2017. Over this decade the manufacturing job reallocation rate fell by 8.71 percentage points. Of this, the within component—the change in reallocation holding the age distribution fixed—accounts for 5.71 percentage points, about two-thirds of the total, with the between and cross components contributing 1.63 and 1.37 points, respectively.[9] The aging of the business sector matters, but the bulk of the decline occurred *within* age groups—exactly the pattern DHJM report for the United States, and one that directs attention to changes in behavior rather than changes in composition.

Table 4 provides further texture by cross-tabulating net job creation, contributions to

[9] That is, the decomposition in equation (10) yields $-8.71 = -5.71 - 1.63 - 1.37$. The aging of the business sector thus contributed to the decline, but it is not the main driver.

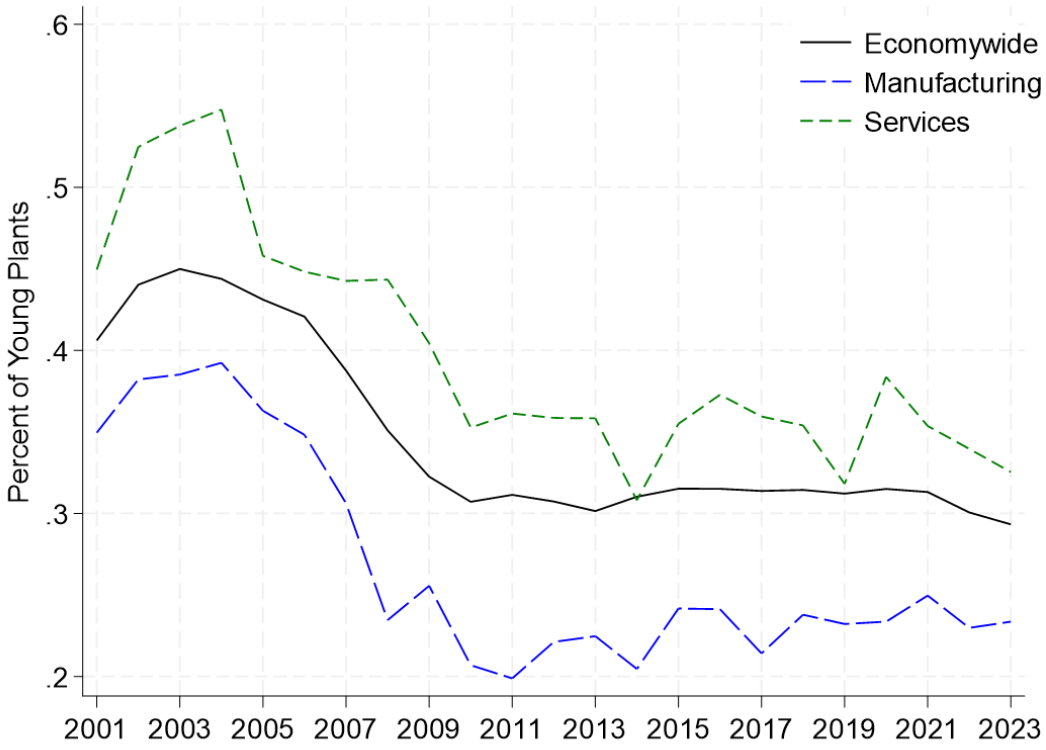


Figure 4: The aging of the Korean business sector

Notes: Authors' calculations from the Census on Establishments microdata (2001–2023). Young establishments are those aged zero to six years, based on the reported founding year. Shares are computed on total employment.

reallocation, and employment shares by age and size over 2001–2023. Three observations stand out. First, the celebrated job creation of small businesses is almost entirely an entry phenomenon: establishments in their birth year add 365,000 jobs per year in the small-size class, while small establishments aged one to six years *destroy* 126,000 jobs per year on net—the signature of up-or-out selection. Second, once entry is set aside, it is large establishments (250 or more employees) that contribute most to net job creation, consistent with the evidence that age, not size, is the relevant margin (Haltiwanger et al., 2013; Cho et al., 2017). Third, and most important for our purposes, mature establishments account for the *largest* share of job reallocation—39.3 percent, exceeding the 38.8 percent of young establishments—so mature businesses were themselves a major carrier of measured dynamism. A purely young-firm account of Korean dynamism sits uneasily with this fact.

Table 4: Job creation, Reallocation, and Employment Shares by Age and Size

	Small	Medium	Large	Total
Panel A. Net job creation (persons, annual average)				
Birth	365,164	47,462	21,940	434,566
Young	−126,239	11,385	12,768	−102,086
Mature	−75,860	−13,626	12,574	−76,912
Total	163,065	45,221	47,282	255,568
Panel B. Contribution to the job reallocation rate (percent)				
Birth	19.0	2.0	0.9	21.9
Young	30.4	5.7	2.7	38.8
Mature	20.3	10.4	8.6	39.3
Total	69.7	18.1	12.2	100.0
Panel C. Employment share (percent)				
Birth	6.8	0.8	0.4	8.0
Young	26.1	5.8	2.8	34.7
Mature	25.5	16.8	15.0	57.3
Total	58.4	23.4	18.2	100.0

Notes: Authors' calculations from the Census on Establishments microdata (2001–2023); entries are annual averages over the sample period. Birth denotes establishments aged zero; Young incumbents, ages one to six; Mature, age seven or more. Small denotes establishments with fewer than 50 employees; Medium, 50–249; Large, 250 or more employees.

6.2 Age-specific responsiveness

We now estimate responsiveness separately by establishment age, interacting productivity with period and with an indicator for mature status (age seven or above):

$$y_{it+1} = \sum_{\tau} \sum_{a \in \{Y,M\}} \beta^{y}_{\tau a}\, z_{it}\, \mathbf{1}\{t \in \tau\}\, \mathbf{1}\{a_{it} = a\} + \sum_{\tau} \sum_{a} \lambda_{\tau a}\, \ln n_{it}\, \mathbf{1}\{t \in \tau\}\, \mathbf{1}\{a_{it} = a\} + \Gamma' \mathbf{c}_{it} + \delta_{s(i)t} + \varepsilon_{it+1}. \tag{11}$$

Size is interacted with period and age throughout, so that a decade-specific change in the size–growth or size–exit relationship is not absorbed into the productivity coefficient. Table 5 reports the results for both margins.

The employment margin. In the 2000s mature establishments were the more responsive group—a fact worth stating on its own, since it is the reverse of the U.S. ordering. Both groups became less responsive thereafter, and here the comparison is complicated by the growth measure itself. Under the standard measure the decline appears concentrated among mature establishments, and significantly so. But by the identity of equation (13), that comparison contains the exit margin, which enters with a coefficient of -2 and whose own age asymmetry runs in the opposite direction. On continuing establishments—the clean measure of employment adjustment—the ordering reverses: young establishments lose almost twice as much responsiveness as mature ones, and the difference between them is precisely estimated. Section 6.3 sets out the arithmetic and reports all three comparisons side by side. On this margin, then, Korea replicates rather than reverses the U.S. pattern: young businesses lost the most adjustment capacity.

The exit margin. The exit margin behaves differently, and this is where Korea departs. In the 2000s, mature establishments were *more* selective than young ones—a low-productivity establishment that had been in operation for some years was markedly more likely to disappear than an equally unproductive young one—and the entire subsequent weakening of the productivity–exit gradient belongs to them. Among young establishments the gradient is statistically unchanged. The gap between the two age groups is large and precisely estimated in the raw linear specification, and it survives placing the two periods on a common scale, where the mature weakening is on the order of two-fifths; Section 6.3 reports the scale-free estimates. Whichever specification is used, the sign and the location of the asymmetry are the same: the productivity–survival link weakened for established businesses and not for young ones.

How robust is the age threshold? Because seven years is a convention inherited from DHJM rather than a structural boundary, we re-estimate the age contrast on the exit margin at cutoffs of five, ten, and fifteen years. Its magnitude is stable across all four thresholds and is largest at the youngest cutoff; statistical significance weakens as the mature group is narrowed and establishments in their teens are reclassified as young, which is a power effect rather than a change in the pattern. Appendix B.1 reports the full profile for both margins.

Table 5: Responsiveness by Age: Young vs. Mature

	Employment growth		Exit	
	(1)	(2)	(3)	(4)
$z_{it}\times$ Young	0.094***		−0.000	
	(0.0080)		(0.0035)	
$z_{it}\times$ Young × trend	−0.001		−0.000	
	(0.0006)		(0.0002)	
$z_{it}\times$ Mature	0.138***		−0.031***	
	(0.0067)		(0.0030)	
$z_{it}\times$ Mature × trend	−0.003***		0.001***	
	(0.0005)		(0.0002)	
$z_{it}\times$ Young × 2000s: β_1^Y		0.090***		−0.000
		(0.0073)		(0.0033)
$z_{it}\times$ Young × 2010s: β_2^Y		0.091***		−0.006***
		(0.0055)		(0.0023)
$z_{it}\times$ Mature × 2000s: β_1^M		0.132***		−0.028***
		(0.0062)		(0.0028)
$z_{it}\times$ Mature × 2010s: β_2^M		0.098***		−0.009***
		(0.0043)		(0.0018)
p-value: $\beta_1^Y = \beta_2^Y$		0.909		0.114
p-value: $\beta_1^M = \beta_2^M$		0.000		0.000
Observations	644,880	480,414	672,366	497,983
Sample	2001–2023	No-crisis	2001–2023	No-crisis

Notes: Authors' estimates from the Mining and Manufacturing Survey and Economic Census microdata, following equation (11). Columns (1)–(2): dependent variable is the DHS employment growth rate; columns (3)–(4): exit indicator (linear probability model). Establishment age is the calendar year minus the founding year, so that an establishment is age zero in its founding year; Young denotes ages zero to six and Mature age seven or more. All specifications include industry, year, and industry-by-year effects, establishment size and age controls, initial employment, and the employment–productivity interaction. The no-crisis sample excludes 2008–2010 and 2020 onward. Standard errors clustered at the establishment level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Because the growth measure embeds exit, columns (1)–(2) are not a clean measure of employment adjustment; Table 6 separates the two margins. Table B1 shows that the pattern is robust to alternative young/mature cutoffs of five, ten, and fifteen years.

The picture that emerges is sharper than the pooled estimates allow, and more useful. Korea's business sector did not become uniformly sclerotic, and its mature establishments did

not simply stop adjusting employment. What changed is that being unproductive stopped predicting exit for establishments that had been in operation for some time.[10]

6.3 Separating the two margins

The two margins estimated so far are not independent, and the connection is algebraic rather than empirical. The symmetric growth rate of Davis et al. (1996) assigns exiting establishments $g_{it} = -2$, so

$$g_{it} \;=\; g^{0}_{it} \;-\; 2\,x_{it}, \qquad g^{0}_{it} \;\equiv\; g_{it} \cdot \mathbf{1}\{\text{continue}\}, \tag{12}$$

and because least squares is linear, provided the three regressions share a sample and a right-hand side the estimated coefficients inherit the identity exactly:

$$\beta^{g} \;=\; \beta^{g0} \;-\; 2\,\beta^{x}. \tag{13}$$

Measured growth responsiveness therefore loads on the exit margin with a coefficient of -2, and in our setting this is not a small matter. Panel A of Table 6 simply displays the identity, period by period and age group by age group. Pooled, nearly two-thirds of the measured decline in β^{g} is contributed by the exit margin rather than by any change in how continuing establishments adjust employment. By age the asymmetry is starker still: for mature establishments the exit margin accounts for four-fifths of the measured decline, whereas for young establishments it works mildly in the opposite direction. The apparently mature-specific weakening of employment growth responsiveness is thus largely the survival margin passing through the measure.

Panel B reports the quantity that actually bears on employment adjustment: the change in β^{g} estimated on continuing establishments only, which drops exiters rather than assigning them a growth rate. This is the row to read for the employment margin, and it reverses the age ordering of Panel A. Both age groups adjust less; young establishments lose almost twice as much as mature ones; and the difference between them is precisely estimated and of the opposite sign to the corresponding difference in Panel A. Responsiveness on the employment margin did fall, and significantly. It fell by substantially less than the headline number implies, and it fell hardest among the young.

Properly measured, then, the pooled Korean facts line up with DHJM. Employment ad-

[10] Appendix B.5 plots the age-specific coefficients on overlapping three-year windows. The gap between mature and young establishments on the exit margin opens gradually over the 2010s rather than at any single date, which is why we do not pursue an event-study design around a specific institutional change.

justment became less responsive to productivity, by roughly a third less than the headline coefficient suggests once the exit content is removed, and exit selectivity weakened once the base rate is accounted for (Section 5.3). Both margins moved, and both moved in the direction the U.S. and European evidence would lead one to expect. The methodological implication is general: growth-margin and exit-margin results should not be presented as independent evidence for the same conclusion, in this literature or elsewhere, because equation (13) makes them mechanically related. What the pooled agreement conceals is that it does not survive disaggregation by age, and that the disagreement is confined to one of the two margins.

A useful check on the decomposition is available at the industry level and uses no standard errors at all. Estimating the responsiveness change separately within each of the 21 two-digit manufacturing industries and simply counting signs, the standard growth measure shows weakening in a large majority of industries for mature establishments and in fewer than half for young ones; measured on continuing establishments the counts turn over, with young establishments weakening in more industries than mature ones. A count-based statistic that uses no standard errors and imposes no functional form reverses in the same direction and at the same point as the regression estimates, which is what one would expect if the reversal is a property of the measure rather than of the estimator or the inference.[11]

Scale-free robustness. Because the exit rate fell substantially, Appendix B.2 checks the age contrast with latent-scale and base-rate-invariant measures. The probit coefficient for mature establishments weakens by about 41 percent, while the corresponding change for young establishments is small and statistically insignificant; the mature–young difference is significant. We treat these estimates as robustness checks rather than as the main specification, but they preserve the central age asymmetry in the linear results.

6.4 A fixed-cohort test: composition and vintage

The mature-specific weakening could in principle be a composition effect. The set of establishments classified as mature is not the same set in the two periods, so a group-level change in selectivity need not reflect any change in behaviour. We address this with a three-way partition that holds cohort membership fixed. *Fixcohort* establishments are those born in 1994 or earlier and are therefore mature in every year of the 2001–2019 window; *newmature* establishments are born after 1994 and cross age seven within the sample; *young* is as before.

[11] Appendix B.3 reports the counts for both margins and both age groups, with exact sign tests. The same exercise establishes that the decline is broad-based across manufacturing rather than the product of one or two sectors.

Table 6: Where the measured decline in growth responsiveness comes from

	Mature	Young	Pooled
Panel A. The accounting identity, $\Delta\beta^g = \Delta\beta^{g0} - 2\,\Delta\beta^x$			
Measured growth margin, $\Delta\beta^g$ (exiters = −2)	−0.0362	−0.0102	−0.0242
	(0.0079)	(0.0091)	(0.0077)
of which: continuing component, $\Delta\beta^{g0}$ (exiters = 0)	−0.0072	−0.0124	−0.0087
of which: exit component, $-2\,\Delta\beta^x$	−0.0290	0.0022	−0.0155
memo: exit component as a share of $\Delta\beta^g$ (%)	80	−21	64
Panel B. The employment-adjustment margin: continuing establishments only			
$\Delta\beta^g$, continuers	−0.0136	−0.0246	−0.0174
	(0.0035)	(0.0040)	(0.0034)
Mature − young	+0.0110	(0.0031), significant at 1 percent	
Panel C. The exit margin, same sample and right-hand side			
$\Delta\beta^x$	0.0145	−0.0011	0.0077
	(0.0035)	(0.0041)	(0.0035)
Mature − young	+0.0156	(0.0030), significant at 1 percent	
Observations (full / continuers)		455,628 / 421,672	

Notes: All entries are 2011–2019 minus 2001–2007; standard errors clustered on the establishment in parentheses. *Panel A is an accounting identity, not three independent findings.* Its three rows use an identical sample and right-hand side, so equation (13) holds exactly within each column: pooled, $-0.0087 - 2(0.0077) = -0.0242$. The memo line expresses the exit component as a percentage of the measured decline; it is negative for young establishments because there the two components have opposite signs, so the exit margin offsets rather than amplifies the measured decline. *Panel B is the estimate to read for employment adjustment.* It drops exiting establishments rather than assigning them a growth rate of zero, which is why the sample is smaller and the estimate differs from $\Delta\beta^{g0}$. The mature-minus-young difference in the measured growth margin of Panel A is −0.0259 (0.0068), significant at 1 percent; it has the opposite sign to the corresponding difference in Panel B, and the gap between the two is the exit content of the measure. Panel C reports the exit margin on the identity sample, so it is directly comparable with Panel A; the scale-free counterparts are in Table B3. The sample differs from Table B2 because the growth equation requires the growth rate to be defined.

The design discriminates among three accounts. A pure composition story is ruled out if fixcohort alone weakens, since fixcohort is a fixed set of establishments tracked over time. A pure vintage story—in which the decline is a property of when an establishment was born—predicts that newmature, a more recent generation, should not weaken. A status-based story—in which what matters is being an established business, whenever founded—predicts that newmature should weaken as well. The last of these is the profile of the survival wedge itself, and Section 7.3 returns to what it implies for the candidate mechanisms.

Table 7 reports the results. Fixcohort establishments—the same businesses observed in both periods—show a significant weakening of exit selectivity, significantly larger than that

Table 7: Composition versus vintage: fixed-cohort estimates on the exit margin

	Linear ($\Delta\beta^x$)		Scale-free ($\Delta\theta$)	
	Estimate	p	Estimate	p
Young	−0.0008	0.841	0.0135	0.572
	(0.0040)		(0.0239)	
Fixcohort	0.0224	<0.001	0.0609	0.025
	(0.0036)		(0.0272)	
Newmature	0.0109	0.012	0.0264	0.364
	(0.0044)		(0.0291)	
Fixcohort − young	0.0232	<0.001	0.0474	0.044
Fixcohort − newmature	0.0115	0.001	0.0345	0.213
Newmature − young	0.0117	0.003	0.0129	0.619
Observations	472,640		472,529	

Notes: Changes are 2011–2019 minus 2001–2007. A positive value denotes a weakening of exit selectivity. Fixcohort establishments are born in 1994 or earlier and are mature in every sample year; newmature establishments are born after 1994 and cross age seven within the sample. Standard errors clustered on the establishment in parentheses. The scale-free columns report probit index coefficients as in Table B3; the sample is marginally smaller because of perfect prediction in a small number of industry-by-year cells.

of young establishments, which again show none. The composition explanation is therefore rejected: the wedge opens *within* a fixed set of establishments, not merely between the groups that happen to occupy the mature category in the two periods. Newmature establishments weaken with the same sign, significantly so in the linear specification and with a point estimate of the same sign but less precision once the scale adjustment is applied. The vintage explanation is therefore weakened but not eliminated: the point estimates are graded, with fixcohort weakening more than newmature, and this ordering is itself significant in the linear specification though not in the probit. We note explicitly that a graded pattern of this kind is also consistent with a continuous age gradient, since fixcohort establishments are mechanically older in years than newmature ones, and this design cannot separate the two.

7 Interpreting the Decline: Possible Mechanisms

The preceding sections established a specific fact: the link between productivity and survival weakened among established businesses and not among young ones, and this is not a composition effect. We call this the *survival wedge*—a growing gap between the exit rate a business's productivity would predict and the exit rate it actually faces. This section asks what could produce it. The reduced-form responsiveness coefficient is a sufficient statistic for

the combined effect of whatever forces flatten the mapping from productivity to adjustment, so what follows narrows the field rather than identifying a cause.

7.1 Three possible channels

The first candidate is a change in market toughness. In models with endogenous markups (Melitz and Ottaviano, 2008; Syverson, 2004), a shift in the competitive environment moves the productivity cutoff for survival and the elasticity of scale to productivity, and it operates on both margins at once. This fits the common component of the decline. It does not naturally generate the age asymmetry, since the baseline versions of these models do not distinguish establishments by age within the same industry and year. Nonetheless, it carries testable cross-sectional implications, though their direction is not settled a priori: models of competitive selection predict that responsiveness held up best where pressure rose most, while models in which shocks raise uncertainty or tighten financial constraints predict the reverse (Dixit, 1989; Bloom, 2009; Gopinath et al., 2017). Section 7.2 tests both.

The second is rising adjustment frictions—employment protection for regular workers, seniority-based pay, and the fixed costs of labor adjustment emphasized by Hopenhayn and Rogerson (1993) and Bloom et al. (2007). This is the mechanism DHJM favour for the United States and it is plausible in Korea. It operates on the employment margin and predicts little on the exit margin. The continuers-only estimates of Section 6.3 show no mature-specific decline on the employment margin, which is precisely where this channel would have to appear; if anything the loss of adjustment capacity there is a young-establishment phenomenon. Whatever produced the Korean pattern, it did not work primarily through the cost of adjusting employment.

The third possibility is that some form of support or forbearance came to sustain establishments increasingly independently of productivity, with access rising with operating history. Possible candidates in the Korean context include small-business policy finance and credit guarantees, relationship lending, creditor forbearance, and employment-retention support. Such a channel would flatten the exit gradient while leaving the adjustment behaviour of continuing establishments largely intact, and would operate as a function of established status rather than founding cohort—which is precisely the profile the fixed-cohort estimates of Section 6.4 trace out. Related mechanisms have been studied under the heading of weakened selection and the prolonged survival of unproductive incumbents (Caballero et al., 2008; Adalet McGowan et al., 2018).

7.2 Industry evidence on competitive pressure

The toughness channel is the one candidate that carries cross-sectional implications, so we test it directly. We estimate equation (11) separately within each two-digit manufacturing industry and relate the resulting long-differences to two families of measures. The first is external: exposure to Chinese competition, which shifts the competitive environment facing Korean producers from outside the industry. The second is internal: the concentration of domestic sales. The distinction matters because the external measures are plausibly exogenous shifters of competitive pressure, while concentration is an equilibrium object determined jointly with the selection process we are studying.

Two conventions govern the reading of what follows. Because a weakening of responsiveness is a fall in the coefficient on the employment margin and a rise toward zero on the exit margin, where the coefficient is negative-valued, a positive slope on the employment margin and a negative slope on the exit margin carry the same substantive meaning. And the direction to test is not obvious in advance. Models of competitive selection (Melitz and Ottaviano, 2008; Syverson, 2004; Asplund and Nocke, 2006; Raith, 2003) predict that tougher competition steepens the mapping from productivity to both scale and survival, so responsiveness should have held up best where pressure rose most. A second family predicts the opposite: shocks that raise uncertainty widen the region of inaction (Dixit, 1989; Bloom, 2009), tightening financial constraints prevent productive establishments from expanding where the shock lands hardest (Midrigan and Xu, 2014; Gopinath et al., 2017), and creditor or policy forbearance responds most visibly to distress, suppressing exit precisely in the industries hit hardest. We report the trade results against both alternatives and label each with the theory it tests.

External pressure: exposure to Chinese competition. The most widely studied source of competitive pressure on Korean manufacturing over this period is the rise of China. Table 8 reports inverse-variance-weighted regressions of the industry long-differences on three measures of exposure, constructed from bilateral trade data for eighteen harmonised manufacturing groups: Chinese import penetration of the Korean market, in the tradition of Autor et al. (2013); displacement, defined as the change in China's share of Korea's own export destinations net of the change in Korea's share of those same destinations; and the change in Korea's destination-market share on its own. Exposure is measured as the change between the 2001–2007 and 2011–2019 averages, matching the window over which the responsiveness differences are estimated.

The competitive-selection account is in difficulty before the cross-section is examined.

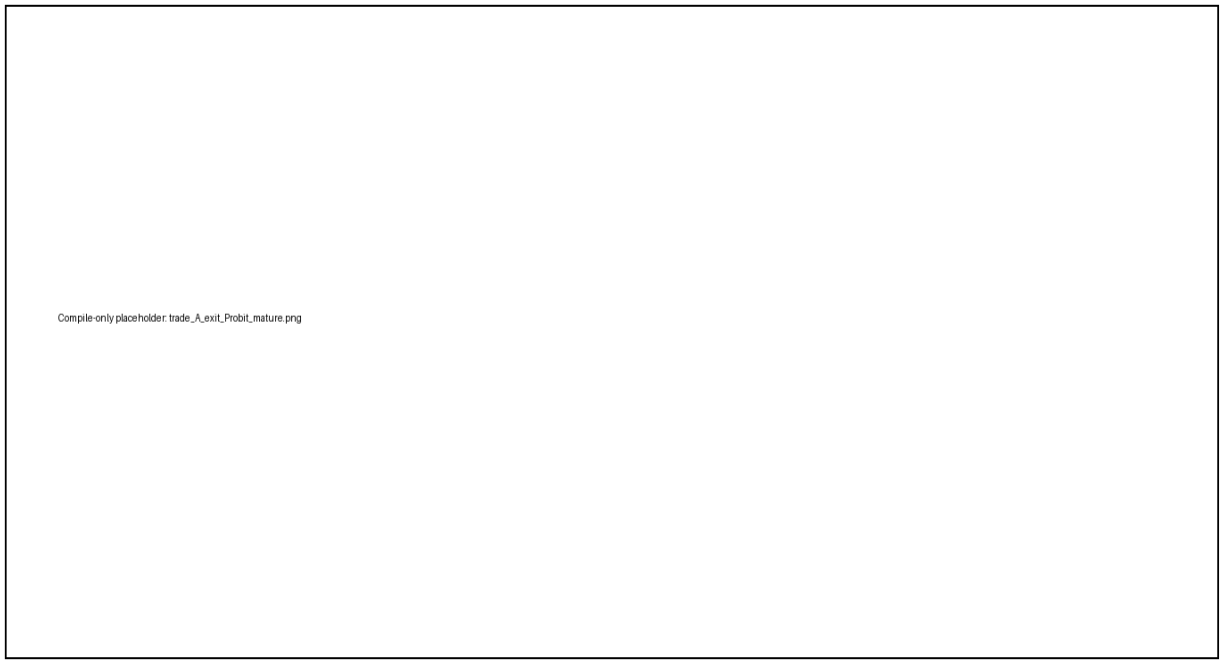


Figure 5: Exit selectivity and exposure to Chinese competition, mature establishments

Notes: Eighteen harmonised manufacturing groups. The two exposure measures are essentially uncorrelated across industries ($r = -0.04$), so the panels carry independent information. Dashed lines are inverse-variance-weighted fits; slopes and significance are reported in Table 8.

Pressure on Korean manufacturing rose over the sample period on every measure we construct, and tougher competition predicts a steeper mapping from productivity to survival, not a flatter one. For this channel to account for what we observe, toughness would have to have fallen. Table 8 asks whether cross-industry heterogeneity can nonetheless rescue it—whether responsiveness held up best where pressure rose most—and finds only weak support. Every point estimate carries the predicted sign, but only two reach the ten percent level, both for mature establishments on the scale-free margin (displacement, $p = 0.07$; destination share, $p = 0.10$). Against the opposite alternative the table is unambiguous: not one cell offers any support for the view that trade distress flattened selection in the industries most exposed to it, with $p \geq 0.60$ throughout.

The age ordering is also worth recording, since it does not reproduce the aggregate pattern. Import penetration is associated less strongly with the change in exit selectivity among mature establishments than among young ones, which is the reverse of the age asymmetry the aggregate estimates require. The export-market measures are ordered as the aggregate pattern would require, with mature coefficients larger in absolute value than young ones in both specifications, but they are mutually correlated, so this is closer to a single observation than to several independent ones.

Table 8: Change in exit selectivity and exposure to Chinese competition

		Linear		Scale-free (probit)	
Exposure measure	Group	Slope	S.E.	Slope	S.E.
Chinese import penetration	Mature	−0.017	(0.066)	−0.231	(0.426)
	Young	−0.041	(0.059)	−0.321	(0.347)
Displacement	Mature	−0.164	(0.119)	−1.227	(0.786)
	Young	−0.100	(0.114)	−0.490	(0.689)
Korea's destination share	Mature	−0.525	(0.604)	−4.934	(3.598)
	Young	−0.334	(0.573)	−2.912	(3.217)

Notes: Each cell is the slope from an inverse-variance weighted regression of the industry-level change in exit selectivity on the industry-level change in trade exposure, across eighteen harmonised manufacturing groups; the concordance from the twenty-one two-digit industries is described in Appendix A.6. Exposure is the change between the 2001–2007 and 2011–2019 averages, and all three measures are signed so that a higher value denotes greater pressure on Korean producers. Because the two families of theory discussed in Section 7.1 predict opposite signs, significance is assessed one-sided against each, with the direction fixed by the mechanism rather than by the estimates. Against the competitive-selection alternative (a negative slope, meaning selection strengthened where pressure rose most), two cells reach the ten percent level: displacement for mature establishments ($p = 0.069$ probit, $p = 0.094$ linear) and Korea's destination share for mature establishments ($p = 0.095$ probit). No other cell reaches ten percent, and the smallest p-value among the young-establishment cells is 0.184. Against the opposing alternative—that uncertainty, financial constraints, or distress-triggered forbearance flattened selection where exposure was greatest—every cell has $p \geq 0.60$. Excluding other transport equipment, the industry with the highest leverage on the destination-share measure, leaves every association statistically indistinguishable from zero on either alternative; unweighted and rank-based (Spearman) versions give the same conclusion. A fourth measure, Korea's comparative advantage relative to China, is reported in Appendix B.8 only: it is correlated with four independent proxies for industry size at $|r| = 0.55$–0.66, which violates the condition under which a pooled cross-industry slope can be read as an average of industry-specific effects.

Internal structure: market concentration. Table 9 reports the same specification with changes in domestic concentration in place of trade exposure. Here we do not pre-specify a direction, and we test two-sided. The reason is that concentration has no agreed competitive interpretation: a rising five-firm share is read as evidence of intensified competition by Autor et al. (2020) and as evidence of rising market power by De Loecker et al. (2020). It is also, unlike trade exposure, an outcome of the selection process itself—under functioning selection a common adverse shock raises concentration by removing the least productive producers—so ΔCR5 and $\Delta\beta$ are joint equilibrium objects rather than treatment and response.

With that caveat, the two margins agree, and they point away from a market-power interpretation. On the employment margin the slope on ΔCR5 is positive and significant: industries where concentration rose are industries where responsiveness weakened *less*. On the exit margin the slope is negative—significantly so in both the linear and the scale-free specification—which, because weakening is positive on this margin, carries the same



Figure 6: Change in employment-growth responsiveness and change in concentration across industries

Notes: Each marker is a two-digit manufacturing industry. Dashed lines are inverse-variance-weighted fits, with slopes reported in each panel. The exit-margin version on the common scale is Figure 7

meaning. Disaggregating by age, the association on the exit margin is significant for mature establishments and not for young ones, but the two coefficients are not significantly different from one another and we do not read the contrast as informative. Results are consistently weaker for ΔHHI than for ΔCR5, which is unsurprising given that HHI is dominated by the top of the size distribution and is noisier at the two-digit level.

What the industry evidence can and cannot establish. Three limits bound the reading of both tables. These are between-industry regressions on twenty-one units, or eighteen for the trade measures, so they cannot test a within-industry comparative static such as Melitz and Ottaviano (2008) directly: if the responsiveness–toughness slope is itself heterogeneous across industries, a cross-industry regression need not recover it. The regressions are cross-sectional long differences with an intercept, so any component of a shock common to all industries is absorbed into that intercept and is not identified; what they can exclude is the *sectoral* channel—differential exposure producing differential decline—and not a nationally common one. And concentration remains an equilibrium outcome throughout.

The conclusion we draw is correspondingly modest, and it is negative in both directions. We find no evidence that rising market power drove the decline in responsiveness, and the

Table 9: Change in responsiveness and change in market concentration

		ΔCR5		ΔHHI	
Margin	Group	Slope	p	Slope	p
Employment, standard measure	Pooled	0.673	0.018	0.664	0.087
	Mature	0.711	0.027	0.598	0.185
	Young	0.649	0.035	0.730	0.063
Employment, continuers only	Pooled	0.235	0.039	0.266	0.082
	Mature	0.161	0.166	0.137	0.388
	Young	0.341	0.008	0.401	0.014
Exit, linear	Pooled	−0.212	0.079	−0.189	0.242
	Mature	−0.264	0.044	−0.221	0.225
	Young	−0.161	0.229	−0.163	0.339
Exit, scale-free (probit)	Pooled	−1.533	0.046	−1.173	0.233
	Mature	−1.904	0.031	−1.387	0.221
	Young	−1.208	0.132	−1.121	0.257

Notes: Each cell is the slope from an inverse-variance weighted regression of the industry-level change in responsiveness on the industry-level change in concentration across 21 two-digit manufacturing industries, weighting by the inverse sampling variance of the industry-specific responsiveness estimate. Concentration measures use the three-digit market definition, aggregated to two-digit industries by sales weights. Because concentration admits no agreed competitive interpretation, p-values are two-sided. A positive slope on the employment margin and a negative slope on the exit margin carry the same substantive meaning: rising concentration is associated with *less* weakening of responsiveness.

sign of the association is the opposite of what that story predicts. We find no evidence that external competitive pressure drove it either: pressure rose while responsiveness fell, which is the wrong sign in the aggregate, and the cross-section offers only marginal support for the competitive-selection reading and none at all for the alternative in which trade distress flattens selection. Taken together with the absence of any dateable institutional break and with the finding that the decline is broad-based across manufacturing rather than sectoral (Appendix B.3), we read the industry-level evidence as pointing toward a common national factor rather than industry-specific competitive pressure.

7.3 Status versus vintage: what the fixed-cohort design rules in

The fixed-cohort estimates of Section 6.4 were introduced to rule out composition. They also discriminate among the channels above, because the three cells differ in *when* their members were founded as well as in their current status.

The relevant comparison is between fixcohort establishments, all founded in 1994 or earlier, and newmature establishments, all founded after 1994 and crossing age seven inside

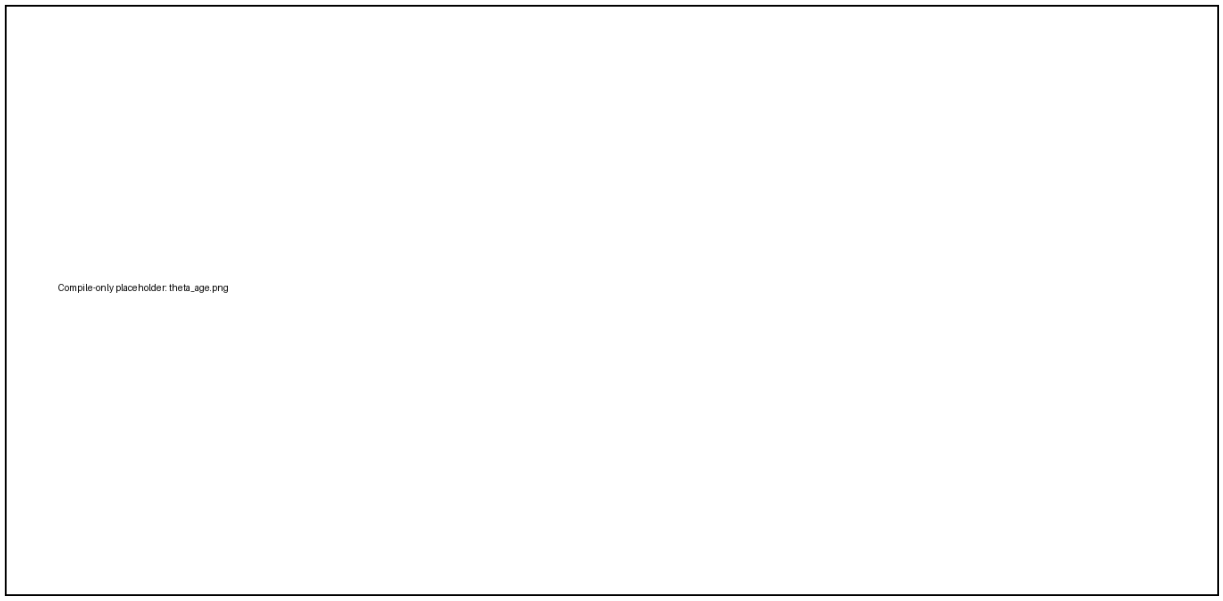


Figure 7: Exit selectivity and market concentration, by establishment age

Notes: A negative slope indicates that industries with rising concentration saw *less* weakening of exit selectivity.

the sample. If the survival wedge were a property of a particular generation—the cohorts that came of age around the 1997 crisis and the restructuring that followed, say, or those formed under the credit conditions of the 1980s—then it should attach to fixcohort and spare newmature, which was founded under different conditions entirely. It does not. Newmature establishments weaken too: significantly in the linear specification, and with the same sign though less precisely estimated on the common scale. Whatever is operating reaches businesses founded a decade apart, once they have been in operation for some years.

This pattern is consistent with a status-based mechanism and weakens a pure vintage interpretation, but it does not eliminate vintage effects. It is consistent with eligibility rules—for policy finance, guarantee renewal, or relationship lending—that key on operating history, track record, or accumulated collateral rather than on founding date. It is harder to reconcile with explanations resting exclusively on the formative conditions of particular cohorts.

Two qualifications limit how far this can be pushed. The point estimates are graded, with fixcohort weakening more than newmature, and this ordering is significant in the linear specification. Because fixcohort establishments are mechanically older in years than newmature ones, a continuous age gradient would produce the same gradation, and this design cannot separate a continuous gradient from a threshold at maturity. And the exercise identifies the profile of the mechanism, not the mechanism itself.

7.4 Institutional candidates and limits of the evidence

The age profile of the decline suggests looking for mechanisms whose incidence rises with operating history. Possible candidates include policy finance, credit guarantees, relationship lending, and other forms of creditor or policy forbearance. Each could, in principle, weaken productivity-based exit selection without producing a mature-specific decline in the employment adjustment of continuing establishments. Related mechanisms have been documented in other settings (Caballero et al., 2008; Adalet McGowan et al., 2018).

Our data do not contain establishment-level measures of guarantee coverage, policy-loan receipt, or lender forbearance, so we cannot identify any of these mechanisms directly. Nor does the rolling evidence reveal a single dateable break that would support a clean event-study design. We therefore treat these channels as possible candidates rather than established explanations. Direct tests require exposure measures constructed independently of exit itself.

7.5 What we can and cannot conclude

The case for the third channel rests substantially on evidence against the first two, and elimination is weaker than identification: any unmodelled force with the same age profile would fit equally well. Three features of the wedge constrain what that force can look like. It acts on the survival margin and not on the employment-adjustment margin of continuing establishments. It acts as a function of established status rather than founding cohort. And it is national rather than sectoral: it shows no systematic relationship with industry exposure to import competition, export displacement, or concentration, and the industry-level nulls are what a common national factor would predict.

What we can state with more confidence is negative and still useful. Korea's distinctive survival-side decline is not driven by young firms and is not a composition effect. The evidence also points away from a mature-specific employment-adjustment friction. A pure vintage explanation is weakened, but not eliminated. What remains is a weakening of the link between productivity and survival among established businesses. Identifying what produced that weakening requires variation in exposure to candidate mechanisms measured independently of exit itself—firm-level records of guarantee coverage or policy-loan receipt, for instance—and we take that up in ongoing work.

8 Aggregate Implications: Counterfactual Analysis

The estimates so far establish that responsiveness declined; this section quantifies how much that decline mattered. Following DHJM, we conduct partial-equilibrium counterfactuals that replace the 2010s responsiveness coefficients with their 2000s values while holding all realized fundamentals—the distributions of productivity and initial employment—at their actual values.

Two features of the exercise follow from Sections 5.3 and 6.3. First, because the linear exit coefficient confounds selectivity with the base rate, a counterfactual built on it restores both at once. We therefore report two bounds. The *inclusive* counterfactual restores the 2000s exit coefficients as estimated, and so implicitly restores the 2000s exit environment as well; the *selectivity-only* counterfactual restores the behavioural parameter while leaving the exit rate at its actual 2010s level. The latter is the conservative bound and the one we emphasize. Second, log employment enters as a control and its own size–growth relationship dominates the productivity signal in predicted values by an order of magnitude; we therefore exclude it from the counterfactual prediction while retaining it as an estimation control, so that the reported quantities are differences attributable to the productivity channel rather than full predicted levels.

Panel A of Table 10 reports the reallocation results. In the estimation sample the manufacturing job reallocation rate averaged 17.3 percent per year over 2001–2007 and 14.3 percent over 2011–2019, a decline of 2.9 percentage points.[12] The selectivity-only counterfactual attributes 1.8 percentage points of that decline to weakened responsiveness across the employment-adjustment and survival margins jointly, about three-fifths. The inclusive counterfactual attributes 3.2 percentage points, slightly more than the entire observed decline; over-attribution of this kind is familiar from DHJM's own exercise and reflects the partial-equilibrium accounting, in which offsetting adjustments in wages, prices, and entry are switched off. Decomposing the selectivity-only figure by margin, the employment-adjustment and exit margins contribute about 0.9 percentage points each. Thus the survival margin accounts for roughly half of the recovered reallocation effect, even though it is the margin that carries the distinctive age asymmetry.

Panel B reports the productivity implications, which follow from the fact that aggregate productivity is an activity-weighted average of establishment productivity, so the allocation of employment growth across the productivity distribution matters for aggregate growth.

[12] The level differs from the economywide series in Figure 1 because the exercise is confined to the manufacturing estimation sample—establishments with ten or more employees for which lagged productivity is observed.

Table 10: Counterfactual analysis: the contribution of declining responsiveness

	Selectivity only	Inclusive
Panel A. Job reallocation rate (percentage points per year)		
Total effect	−1.78	−3.21
growth margin	−0.86	—
exit margin	−0.92	—
Memo: observed decline	−2.91	
Panel B. Aggregate productivity growth (percentage points per year)		
Labor productivity	−0.88	−1.58
Total factor productivity	−0.30	−0.44

Notes: Entries are differences between predictions under the actual estimated coefficients and predictions in which the 2011–2019 responsiveness coefficients are replaced by their 2001–2007 values, holding the realized distributions of productivity and initial employment fixed. The selectivity-only column restores the scale-free selectivity parameter while leaving the exit rate at its actual level; the inclusive column restores the estimated linear coefficients and therefore also restores the 2000s exit environment. Log employment is excluded from the counterfactual prediction and retained as an estimation control; see the text. Panel B is computed for aggregate log labor productivity and log TFPS respectively.

The selectivity-only counterfactual raises annual aggregate labor productivity growth by 0.88 percentage points and aggregate TFP growth by 0.30 percentage points; the inclusive counterfactual raises them by 1.58 and 0.44 percentage points respectively. Set against the observed slowdown in annual labor productivity growth in this sample, the conservative bound accounts for roughly one-seventh and the inclusive bound for roughly one-quarter.

Three caveats bound the interpretation. First, the exercise is a partial-equilibrium accounting device in the spirit of DHJM: wages and prices are held fixed, so the general-equilibrium feedbacks emphasized by Hopenhayn and Rogerson (1993) are absent. Second, the counterfactual holds the realized shock distributions and the entry margin at their actual values; it does not model how entry or the shock process would themselves respond to a more flexible environment. Third, the predictions inherit the specification of the responsiveness regressions. We therefore read these numbers as disciplined magnitudes—evidence that the responsiveness channel is first-order—rather than as precise welfare statements.

9 Conclusion

Using panel data covering the universe of Korean establishments over 2001–2023, this paper has documented a decline of roughly 28 percentage points in the job reallocation rate and traced its sources within the framework of Decker et al. (2020).

The decline is broad-based across manufacturing and services and predates the pandemic.

Productivity dispersion did not systematically decline, estimated innovation dispersion fell modestly, and measured persistence rose. We therefore view the shock evidence as mixed but insufficient to explain the pronounced decline in reallocation on its own. By contrast, the mapping from productivity to adjustment and survival became flatter.

Where that reduced response sits took some care to establish, because the standard tools obscure it in two ways. The symmetric growth rate assigns exiting establishments a growth rate of -2, so growth-margin and exit-margin responsiveness are linked by an exact identity, and most of the measured decline in growth responsiveness in Korea is the survival margin passing through the measure; on continuing establishments Korea resembles the United States, with both age groups adjusting less and young establishments considerably less. Symmetric care is needed on the exit margin, where the slope of a linear probability model scales with the exit rate and the Korean exit rate more than halved; placed on a common scale, the weakening of selectivity is real but smaller than the raw coefficients imply. Neither adjustment is peculiar to Korea, and both cut against our own earlier estimates.

What is distinctive about Korea survives both. The weakening of the link between productivity and survival belongs to mature establishments—the group that selected most sharply on productivity to begin with—and not to young ones. It is not a composition effect: it appears within a fixed cohort of businesses that were mature throughout the sample, and it reaches businesses founded a decade apart once they have an operating history. Korea's dynamism did not decline because the economy ran out of young firms. It declined because being unproductive stopped predicting exit for firms that had been in operation for some years, at exactly the point when those firms' share of employment was rising.

The aggregate stakes are large. Holding responsiveness on both the employment-adjustment and survival margins at their 2000s levels accounts for about three-fifths of the observed decline in manufacturing job reallocation and for close to a percentage point per year of labor productivity growth. The survival margin contributes about half of the reallocation effect. Weakened selection among incumbents is, on these numbers, a first-order component of Korea's growth slowdown rather than a micro-level curiosity.

For policy, the implication is a reordering of priorities. An entry-promotion agenda addresses a margin on which Korea does not look unusual. The margin on which it does look unusual is selection among incumbents, where the relevant levers—the design of small-business credit support, the treatment of persistently unprofitable firms, and the terms on which established businesses can be wound down—are different instruments with different constituencies.

The mechanism behind the wedge remains open. Our evidence points away from adjust-

ment frictions and away from age-neutral changes in competitive toughness, and leaves open the possibility of mechanisms that sustain established businesses independently of productivity and whose incidence rises with operating history. Testing that directly requires exposure measures constructed outside the exit data themselves—firm-level records of guarantee coverage or policy-loan receipt, for instance—and we regard the identification of those drivers, rather than the measurement of the decline, as the binding constraint on policy conclusions. It is also a question that will not remain Korean for long. Business populations are aging across the advanced economies, and the Korean experience suggests that what matters is less how many young firms an economy has than whether its old ones still face the discipline of exit.

A Data Appendix

This appendix provides the data-source and sample-construction details omitted from the main text. The baseline establishment analysis combines the universe *Census on Establishments* with manufacturing production surveys; the auxiliary firm-level analysis uses the *Survey of Business Activities*. The goal of the construction is to measure entry and exit from a universal business frame while using production data only where they are available.

A.1 Data sources and coverage

Our primary source is the *Census on Establishments*, an annual complete count of establishments in Korea. We use the 2000–2023 waves. Establishment identifiers are not available before 2000, which fixes the start of the panel. The census provides establishment identifiers, employment, founding year, and industry classification and is the population frame for the establishment panel. The universal coverage is essential for the entry and exit measures: an establishment can disappear from a production survey because it falls below a survey threshold, whereas disappearance from the census panel indicates that it is no longer observed in the population frame.

Production variables needed to construct establishment productivity come from the *Mining and Manufacturing Survey* and the quinquennial *Economic Census*. The Mining and Manufacturing Survey covers manufacturing establishments with ten or more employees. The microdata used in this paper are available for 2001–2009, 2011–2014, 2016–2019, and 2023. Economic Census observations are used for 2010, 2015, and 2020. The resulting production panel is therefore not a balanced annual panel over 2001–2023: production-based regressions use years for which the required microdata are available, and we do not interpolate missing establishment-level production records. In particular, 2021–2022 production microdata are not used in the baseline productivity regressions.

Two auxiliary sources are used. First, the business register, the population frame of enterprises, is merged for 2005–2020 to obtain firm-level age and size for multi-establishment firms. Second, the *Survey of Business Activities* (2006–2023) provides an external-validity check at the firm level for both manufacturing and services. This survey covers firms with fifty or more regular employees, so it is not used to characterize small firms or to construct the baseline establishment results.

Table A1: Data sources and their role in the analysis

Source	Years used	Unit / coverage	Main use
Census on Establishments	2000–2023	Establishments; universe	Employment, founding year, industry, panel identifiers, entry and exit, economywide dynamism
Mining and Manufacturing Survey	2001–2009, 2011–2014, 2016–2019, 2023	Manufacturing establishments with 10+ employees	Sales, intermediate inputs, tangible assets, payroll; baseline productivity regressions
Economic Census	2010, 2015, 2020	Establishments	Production variables in census years; combined with the manufacturing survey
Business register	2005–2020	Firms / enterprises	Firm age and size for multi-establishment firms
Survey of Business Activities	2006–2023	Firms with 50+ regular employees	Firm-level robustness and services evidence using labor productivity

Notes: Years refer to the microdata used in this paper. The manufacturing productivity sample is limited by the availability and coverage of production microdata, whereas entry and exit are defined from the universal Census on Establishments panel.

A.2 Sample construction and sector coverage

The business-dynamism series focus on the private sector and cover manufacturing and services. We exclude construction; finance, insurance, and real estate; electricity, gas, steam, and air-conditioning supply; and public administration. Establishments with only unpaid workers are also excluded. Because the production survey is restricted to establishments with at least ten employees, the manufacturing estimation sample is narrower than the universe used to construct the aggregate reallocation, entry, and exit series. This difference is also why the level of job reallocation in the counterfactual sample is lower than the economywide series in Figure 1.

Entry and exit are determined from the census identifiers rather than from presence in the Mining and Manufacturing Survey. An establishment is an entrant when it newly appears in the population panel and an exiter when it is present in year t but absent in $t + 1$. This convention prevents a fall below the ten-employee survey threshold from being misclassified as exit. The 2020 census expanded the population frame; to reduce the resulting discontinuity, the 2020 entry rate is replaced by interpolation in the aggregate entry series. No such interpolation is used for the productivity regressions.

A.3 Panel linkage and the 2007 identifier change

The establishment identification system in the Mining and Manufacturing Survey changed in 2007. As a result, pre-2007 production records cannot always be linked perfectly to Census on Establishments identifiers, and establishments that exited early in the sample are the most likely to be affected. The exit indicator itself is nevertheless defined from the Census on Establishments throughout, not from disappearance from the production survey. The linkage break therefore does not redefine what counts as exit, but it can affect which pre-2007 exiters have usable production records and hence enter the productivity-responsiveness estimation sample. We treat this as a sample-linkage limitation rather than as an exit-measurement problem.

The no-crisis comparisons used throughout the main analysis are 2001–2007 versus 2011–2019. This choice excludes 2008–2010 and the pandemic period beginning in 2020. It also avoids interpreting the immediate global-financial- crisis years as part of the longer-run responsiveness change.

A.4 Age, size, and timing conventions

Establishment age is constructed from the reported founding year as calendar year minus founding year, so an establishment is age zero in its founding year. The baseline age split defines *young* establishments as ages zero to six and *mature* establishments as age seven or more. In Table 4, birth establishments (age zero) are displayed separately, so "Young incumbents" refers only to ages one to six. Alternative cutoffs of five, ten, and fifteen years are reported in Table B1. The fixed-cohort exercise defines *fixcohort* establishments as those born in 1994 or earlier, so they are mature throughout the 2001–2019 comparison window, and *newmature* establishments as those born after 1994 that cross age seven within the sample.

Size classes are defined consistently throughout the paper: Small has fewer than 50 employees, Medium has 50–249 employees, and Large has 250 or more employees. Employment and productivity are measured at the start of the growth interval whenever they enter the responsiveness regressions; the dependent employment-growth and exit outcomes refer to the transition from t to $t+1$.

A.5 Productivity variables

The baseline productivity variable, TFPS, is revenue-based establishment TFP. It is constructed from establishment revenue and inputs using factor revenue elasticities measured by expenditure shares, as described in equation (5). Because TFPS is revenue based, it combines technical efficiency with idiosyncratic demand and price variation. This is intentional: the responsiveness exercise asks how employment and survival react to the profitability signal facing the establishment, not only to physical productivity.

We use two alternative productivity measures. TFPP is a production-function TFP estimate based on the control-function approach of Levinsohn and Petrin (2003) and Ackerberg et al. (2015). Labor productivity (LP) is real value added per worker. LP is transparent but places employment in the denominator and therefore is not treated as an exogenous shock measure; the age-specific LP results are reported as robustness evidence. Each productivity measure is trimmed at the 0.1st and 99.9th percentiles before estimation.

For the shock exercise, productivity dispersion is reported both in raw form and after removing two- and three-digit industry-year means. Persistence is estimated from period-specific AR(1) regressions. As an additional diagnostic, the standard deviation of the estimated innovation falls from 0.219 to 0.203 for two-digit-within-industry TFPS between the pre- and post-2010 periods, with a similar pattern at the three-digit level; for TFPP it falls

from 0.403 to 0.365. These modest declines are why the main text characterizes the shock evidence as mixed rather than as a sharp rejection.

A.6 Industry definitions and cross-industry exercises

The baseline regressions use industry and industry-by-year controls, and the dispersion exercises remove two- or three-digit industry-year means. The industry decomposition is reported for 21 two-digit manufacturing industries. For the trade-exposure exercise, these industries are harmonized to 18 manufacturing groups so that Korean production industries can be matched to bilateral trade measures. The concordance used for this harmonization is available from the authors. Concentration measures are constructed at the three-digit market level and aggregated to two-digit industries by sales weights, as described in Table 9.

A.7 Main data limitations

Four limitations are worth keeping in view. First, production-based responsiveness estimates apply to the manufacturing establishments for which production microdata can be linked, not to the universe used for the aggregate dynamism series. Second, the ten-employee threshold in the Mining and Manufacturing Survey makes the production sample selective with respect to small establishments, although entry and exit themselves are measured from the universal census frame. Third, the 2007 identifier change creates imperfect linkage for some pre-2007 production records, especially for early exiters. Fourth, the Survey of Business Activities excludes firms with fewer than fifty regular employees and therefore serves only as an auxiliary external-validity sample. These coverage differences are the reason we place the most weight on results that are stable across the Census-based dynamism measures, the manufacturing production sample, and the auxiliary firm-level evidence.

B Robustness and Supplementary Results

B.1 Alternative young/mature age cutoffs

Table B1 re-estimates the age-specific decade specification of Table 5 using young/mature thresholds of five, seven (the baseline), ten, and fifteen years, on the no-crisis sample. The mature responsiveness decline is significant at every cutoff on both the employment-growth and the exit margin, and young responsiveness is flat at every cutoff except the youngest, where the exit change is marginally significant. The qualitative pattern—decline concen-

trated in mature establishments—does not depend on the choice of threshold.

Table B1: Age-specific responsiveness under alternative young/mature cutoffs

Cutoff (years)	5	7	10	15
Panel A. Employment growth				
β^Y, 2000s	0.078***	0.090***	0.104***	0.109***
β^Y, 2010s	0.088***	0.091***	0.097***	0.097***
p: $\beta_1^Y = \beta_2^Y$	0.294	0.909	0.375	0.112
β^M, 2000s	0.129***	0.132***	0.131***	0.135***
β^M, 2010s	0.098***	0.098***	0.095***	0.095***
p: $\beta_1^M = \beta_2^M$	0.000	0.000	0.000	0.000
Panel B. Exit				
β^Y, 2000s	0.007*	−0.000	−0.008***	−0.011***
β^Y, 2010s	−0.006**	−0.006***	−0.008***	−0.008***
p: $\beta_1^Y = \beta_2^Y$	0.004	0.114	0.946	0.312
β^M, 2000s	−0.026***	−0.028***	−0.029***	−0.033***
β^M, 2010s	−0.009***	−0.009***	−0.009***	−0.009***
p: $\beta_1^M = \beta_2^M$	0.000	0.000	0.000	0.000

Notes: Decade specification on the no-crisis sample (2001–2007 and 2011–2019), following equation (11), with young/mature thresholds as indicated in the column heads. Cutoff seven is the baseline reported in Table 5. Coefficients are the productivity semi-elasticities of employment growth (Panel A) and of exit (Panel B). All specifications include the full control set. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

B.2 Comparing exit slopes across periods: the base-rate scaling problem

Section 5.3 reports that the raw linear exit gradient fell by roughly half between the two periods and asserts that most of this is mechanical. This appendix sets out the argument and implements three ways of putting the two periods on a common scale.

The scaling problem. The linear probability slope is not invariant to the level of the outcome. Under a latent-index representation $x^*_{it} = \theta_\tau z_{it} + w'_{it}\gamma + u_{it}$ with $u \sim N(0,1)$, the best linear approximation to the conditional exit probability has slope

$$\beta^x_\tau \;=\; \mathbb{E}[\theta_\tau\, \phi(\text{index}_{it})] \;\approx\; \theta_\tau\, \phi\big(\Phi^{-1}(\bar{x}_\tau)\big)\,, \tag{14}$$

where $\bar{x}_\tau$ is the period exit rate, ϕ and Φ are the standard normal density and distribution function, and θ_τ is the scale-free selectivity parameter of interest. The factor $\phi(\Phi^{-1}(\bar{x}))$

depends only on the base rate and falls as exit becomes rarer. Taking logs,

$$\underbrace{\Delta \ln |\beta^x|}_{\text{observed}} = \underbrace{\Delta \ln \phi\big(\Phi^{-1}(\bar{x})\big)}_{\text{mechanical}} + \underbrace{\Delta \ln \theta}_{\text{behavioural}} . \tag{15}$$

Panel A of Table B2 reports the inputs. The pooled exit rate fell from 11.2 to 4.7 percent and the scale factor from 0.190 to 0.099. The observed pooled change is -0.76 log points against a mechanical component of -0.66, so the raw coefficient overstates the behavioural change by a wide margin, and overstates it most where the exit rate fell most.

Three ways of putting the periods on a common scale. The first is the analytic rescaling implied by (15). The second is a probit on the same sample and controls: its index coefficient *is* θ and lies on precisely the latent scale assumed in (14), which makes the two numerically comparable rather than merely directionally so. The analytic log decomposition leaves $0.104/0.762 = 13.6$ percent—about one-seventh—of the pooled change as behavioural, while the corresponding average-marginal-effect decomposition attributes 22.6 percent, about one-quarter, to selectivity. We therefore describe the pooled behavioural share as roughly one-seventh to one-quarter across these decompositions. The age-specific probit estimates in Table B3 are used as a robustness check on the mature–young contrast rather than as the paper's main specification.

The third is non-parametric and uses no index assumption at all. Sorting establishments into productivity deciles and computing each decile's exit hazard relative to the period average removes the base rate by construction. Table B4 reports the profile. In the first period an establishment in the bottom productivity decile faced an exit hazard 2.26 times the average; by the second period this had fallen to 1.74. Fitting a line through the ten decile points, the profile slope flattens from -0.113 ($p = 0.006$) to -0.073 ($p = 0.025$), a decline of 35 percent. Because this relative-hazard statistic is not a decomposition of the LPM coefficient, we use it only as non-parametric corroboration that the productivity–exit relationship flattened, not as an estimate of the behavioural share of the raw coefficient change.

A further check, reported for completeness, is the area under the receiver operating characteristic curve for predicting exit from productivity alone, which falls from 0.594 to 0.569 pooled. This statistic is also invariant to the base rate but conditions on nothing, so we treat it as corroboration only. A complementary-log-log specification yields the same sign for the mature–young difference; its coefficients lie on an extreme-value rather than a normal latent scale and are therefore not commensurable with the analytic rescaling in

Table B2: Exit rates and the mechanical component of the slope change

	Pooled	Mature	Young
Exit rate, 2001–2007	0.112	0.088	0.144
Exit rate, 2011–2019	0.047	0.035	0.076
$\phi(\Phi^{-1}(\bar{x}))$, 2001–2007	0.190	0.160	0.227
$\phi(\Phi^{-1}(\bar{x}))$, 2011–2019	0.099	0.078	0.143
Mechanical $\Delta \ln \lvert\beta^x\rvert$	−0.658	−0.720	−0.465
Observed $\Delta \ln \lvert\beta^x\rvert$	−0.762	—	—

Notes: Estimation-sample exit rates, the implied linear-probability scale factor, and the mechanical component of the change in the estimated slope implied by equation (15).

Table B3: Scale-free exit selectivity by establishment age

	Mature	Young
θ, 2001–2007	−0.1200	−0.0920
	(0.0173)	(0.0173)
θ, 2011–2019	−0.0713	−0.0830
	(0.0175)	(0.0181)
Change	0.0487	0.0089
	(0.0236)	(0.0239)
p-value	0.039	0.708
Difference, mature − young	0.0398	(0.0195), $p = 0.041$
Observations	472,529	

Notes: Probit index coefficients from the age-specific specification with period-interacted productivity and log employment, age- and size-group indicators, and industry-by-year indicators; standard errors clustered on the establishment in parentheses. Index coefficients are not marginal effects and are not directly comparable in magnitude to the linear estimates in Table 5. The mature coefficient weakens by 0.0487/0.1200 = 40.6 percent, whereas the change for young establishments is small and statistically insignificant. These estimates are reported as a robustness check on the age contrast. The pooled analytic and average-marginal-effect decompositions imply behavioural shares ranging from about one-seventh to one-quarter; see the surrounding text.

equation (15), which is why the probit is our primary scale-free specification.

B.3 Breadth of the decline across two-digit industries

Two questions motivate this exercise. First, whether the decline in responsiveness is broad-based or the product of a few sectors—a check that the main text invokes at several points but does not need to display. Second, whether the reversal of the age ordering between the standard growth measure and the continuers-only measure, documented in Section 6.3, appears in a statistic that uses no standard errors and imposes no functional form. We

Table B4: Relative exit hazard by productivity decile

Productivity decile	2001–2007	2011–2019
1 (lowest)	2.261	1.736
2	1.498	1.076
3	1.246	0.955
4	1.058	0.814
5	1.009	0.753
6	0.969	0.727
7	0.939	0.708
8	0.863	0.714
9	0.896	0.687
10 (highest)	0.910	0.872
Profile slope	−0.113	−0.073
	(0.031)	(0.027)
p-value	0.006	0.025

Notes: Each entry is the exit hazard of establishments in that productivity decile divided by the period-average exit hazard, so the statistic is invariant to the level of the exit rate by construction. The profile slope is from an ordinary least squares fit through the ten decile points, with the standard error computed across those points.

estimate equation (11) separately within each of the 21 two-digit manufacturing industries and count the industries in which responsiveness weakened, which on the employment margin is a fall in β^g and on the exit margin a rise in β^x toward zero.

Table B5 reports the counts. On the employment margin measured by the standard growth rate, weakening is widespread among mature establishments and absent among young ones. Measured on continuing establishments the asymmetry reverses, with young establishments weakening in more industries than mature ones. This is the industry-level counterpart of the reversal in Table 6, and it is reassuring that a count-based statistic reproduces it.

On the exit margin the raw linear estimates show the mature-specific pattern clearly—mature establishments weaken in sixteen of twenty-one industries against eight for young ones—but this does not survive the scale adjustment. In the probit specification neither count is distinguishable from chance. We regard this as a power limitation rather than a contradiction: each industry-specific probit is estimated on a fraction of the sample, the scale-free selectivity change is roughly a third the size of the raw one, and a sign test across twenty-one units discards all magnitude information. The pooled and age-split estimates of Table B3, which use the full sample and retain magnitudes, remain the basis for the age result. What the breadth exercise supports is the weaker claim it was introduced to support: the decline is not the product of one or two sectors, and so is not a sectoral effect

Table B5: Breadth of the responsiveness decline across two-digit industries

Margin	Group	Industries weakening	of 21	Sign-test p
Employment, standard measure	Pooled	14	21	0.189
	Mature	16	21	0.027
	Young	10	21	1.000
Employment, continuers only	Pooled	15	21	0.078
	Mature	15	21	0.078
	Young	16	21	0.027
Exit, linear	Pooled	13	21	0.383
	Mature	16	21	0.027
	Young	8	21	0.383
Exit, scale-free (probit)	Pooled	11	21	1.000
	Mature	12	21	0.664
	Young	9	21	0.664

Notes: Each row counts the two-digit manufacturing industries in which responsiveness weakened between 2001–2007 and 2011–2019, from industry-specific estimates of equation (11). Weakening is a fall in the coefficient on the employment margin and a rise toward zero on the exit margin, where the coefficient is negative-valued. The sign test is a two-sided exact binomial test against $p = 0.5$; we report two-sided p-values throughout because the direction of weakening differs across margins, which makes a one-sided test ambiguous to state.

masquerading as an aggregate one.

B.4 Beyond manufacturing establishments: firm-level evidence and services

Is weakening responsiveness specific to manufacturing establishments? Using the firm-level Survey of Business Activities (2006–2023) and labor productivity as the productivity measure, we re-estimate equation (8) for all industries, manufacturing, and services. Table B6 shows the same pattern throughout: responsiveness declines significantly over time in every sample, whether modeled as a linear trend (about -0.003 per year) or as a post-2010 shift. The post-2010 decline is larger in manufacturing (-0.044) than in services (-0.027), though the comparison should be read with the caveat that the survey excludes firms with fewer than fifty employees, and small businesses account for a larger share of services. The responsiveness hypothesis is thus not an artifact of one dataset, one productivity measure, or one sector: it holds at the establishment and firm levels, in manufacturing and in services.

Table B6: Firm-level Employment Responsiveness

	(1) All	(2) Manuf.	(3) Services	(4) All	(5) Manuf.	(6) Services
z_{it}	0.114***	0.111***	0.117***	0.118***	0.123***	0.115***
	(0.0038)	(0.0052)	(0.0057)	(0.0044)	(0.0060)	(0.0070)
$z_{it}\times$ trend	−0.003***	−0.003***	−0.003***			
	(0.0004)	(0.0005)	(0.0005)			
$z_{it}\times$ Post-2010				−0.035***	−0.044***	−0.027***
				(0.0048)	(0.0064)	(0.0073)
Observations	173,994	93,608	80,386	173,994	93,608	80,386
Sample	2006–2023					

Notes: Authors' estimates from the Survey of Business Activities microdata (2006–2023), which covers firms with fifty or more regular employees. The productivity measure is firm-level labor productivity (real sales per worker); the dependent variable is the DHS employment growth rate. "All" comprises manufacturing and services. Controls as in equation (8). Standard errors clustered at the firm-level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

B.5 Three-year rolling estimates

The period comparisons in the main text impose a single break between the 2000s and the 2010s. This appendix reports the same coefficients estimated on overlapping three-year windows, which imposes no break at all and shows whether the change is a level shift, a trend, or an artefact of where the break is placed. Windows are centred on each year from 2002 to 2022 and estimated on the same specification as equation (11), with industry-by-year effects and standard errors clustered on the establishment.

Figure B1 plots the trajectories. Their purpose is to check that the between-period contrast in Tables 2 and 3 does not depend on where the break is placed. It does not. The employment-margin coefficient is lower in every window centred in the 2010s than in any window centred in the early 2000s, and the exit-margin coefficient is likewise closer to zero throughout the later decade. No choice of break year within the sample would reverse either sign, and neither result is generated by the years we exclude.

Two cautions apply. Overlapping windows induce serial dependence across adjacent points, so the trajectories should be read for level rather than for the significance of any individual window; and three-year windows are short enough that the exit-margin estimates, which depend on a relatively rare event, are visibly noisier than the employment-margin ones.

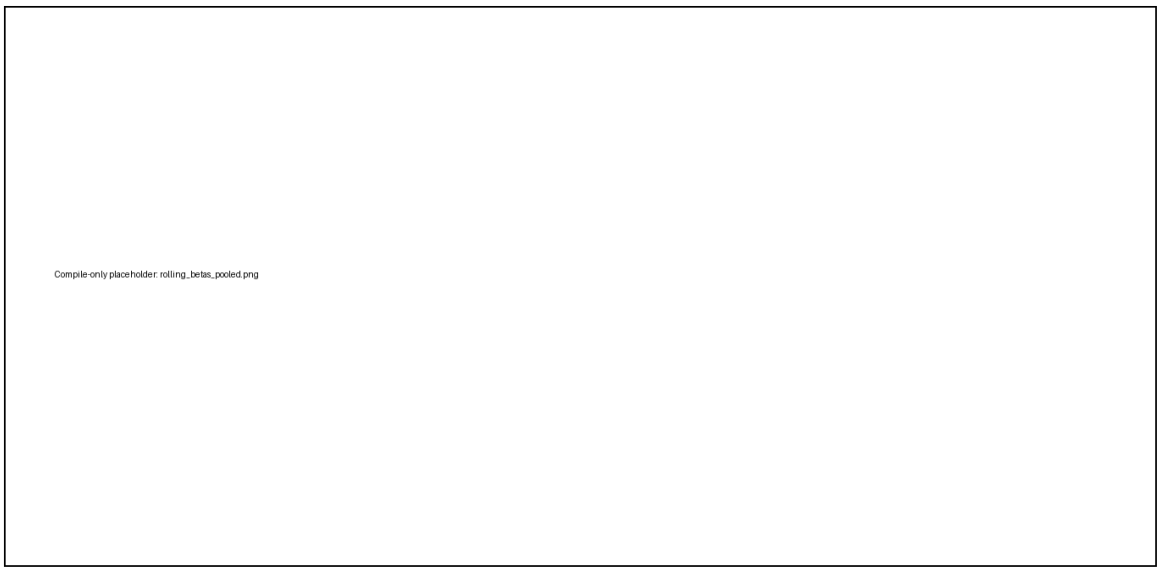


Figure B1: Three-year rolling window estimates of responsiveness

Notes: Windows overlap, so adjacent points are not independent.

B.6 Alternative productivity measures

Tables B7 and B8 reproduce the aggregate and age-specific responsiveness estimates using production-function TFP (TFPP) and real labor productivity (LP) in place of the baseline TFPS. The aggregate decline is robust across all three measures on both margins (Table B7). The age-specific results (Table B8) are robust under TFPP, which preserves both the level ordering (mature more responsive than young) and the mature-specific decline. Under LP the level ordering reverses—young establishments appear more responsive than mature in the 2000s—and young responsiveness also declines on the growth margin. This reversal is a known artifact of measuring productivity as output per worker, which places employment in the denominator and induces a mechanical, age-varying correlation between the regressor and the outcome (division bias); it is most pronounced for young establishments, whose employment is most volatile. We therefore treat TFPS and TFPP as the reliable measures and report LP for transparency. What survives across all three measures, and is the basis for our headline claims, is the asymmetry of the decline—always larger for mature establishments—and the collapse of mature exit responsiveness.

B.7 Industry decomposition of the mature responsiveness decline

Table B9 reports the industry-level change in mature responsiveness, $\Delta\beta^M$, for each two-digit manufacturing industry (three-digit within-industry specification), together with each

Table B7: Responsiveness under alternative productivity measures

	TFPP		LP	
	Growth	Exit	Growth	Exit
2000s	0.077***	−0.013***	0.112***	−0.030***
2010s	0.066***	−0.007***	0.082***	−0.014***
2020s	0.067***	−0.007***	0.076***	−0.009***
p: 2000s = 2010s	0.002	0.000	0.000	0.000
Observations	627,531	653,116	657,435	685,638

Notes: Decade specification for the pooled sample of manufacturing establishments (equation (8)). "Growth" is the productivity semi-elasticity of DHS employment growth; "Exit" the gradient of the exit indicator. TFPP is production-function TFP estimated following Ackerberg et al. (2015); LP is real value-added per worker. Full control set included. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B8: Age-specific responsiveness under alternative productivity measures

	TFPP		LP	
	Growth	Exit	Growth	Exit
β^Y, 2000s	0.077***	−0.013***	0.128***	−0.038***
β^Y, 2010s	0.074***	−0.014***	0.102***	−0.026***
p: $\beta^Y_1 = \beta^Y_2$	0.481	0.626	0.000	0.000
β^M, 2000s	0.085***	−0.016***	0.114***	−0.029***
β^M, 2010s	0.065***	−0.005***	0.074***	−0.008***
p: $\beta^M_1 = \beta^M_2$	0.000	0.000	0.000	0.000

Notes: Decade specification on the no-crisis sample (equation (11)), young/mature cutoff of seven years. Columns as in Table B7. Under LP the 2000s level ordering reverses relative to TFPS and TFPP (see text); the mature-specific magnitude of the decline and the collapse of mature exit responsiveness are common to all three measures. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

industry's employment share and its contribution to the aggregate mature decline. The decline is broad—negative in fifteen of twenty-one industries—and concentrated in mature heavy manufacturing: food and beverages, machinery, electrical equipment, and electronics make the largest negative contributions. Rubber and plastics and other transport equipment provide the largest positive offsets.

Table B9: Industry-level change in mature responsiveness, 2000s to 2010s

Industry	$\Delta\beta^M$	s.e.	Emp. share (%)	Contribution (p.p.)
Food & beverage	-0.134^{***}	(0.023)	7.8	-1.054
Machinery	-0.087^{***}	(0.024)	11.1	-0.965
Electrical eq.	-0.129^{***}	(0.039)	6.7	-0.864
Electronics	-0.039	(0.034)	14.6	-0.567
Basic metal	-0.049	(0.047)	5.1	-0.247
Vehicle	-0.012	(0.040)	12.0	-0.149
Fabricated metal	-0.021	(0.032)	7.1	-0.147
Chemical	-0.020	(0.028)	5.2	-0.104
Precisions	-0.037	(0.047)	2.7	-0.097
Non-metallic min.	-0.025	(0.027)	3.5	-0.088
Apparel	-0.049	(0.045)	1.7	-0.084
Furniture	-0.058	(0.076)	1.4	-0.080
Other mfg.	-0.063	(0.070)	0.9	-0.059
Pulp	-0.016	(0.052)	2.2	-0.034
Printing	-0.027	(0.182)	0.1	-0.002
Wood	$+0.029$	(0.087)	0.7	$+0.019$
Leather	$+0.088$	(0.087)	0.7	$+0.059$
Textile	$+0.066$	(0.044)	2.5	$+0.162$
Pharmaceutical	$+0.100$	(0.082)	1.7	$+0.171$
Other transport eq.	$+0.149$	(0.115)	4.1	$+0.606$
Rubber & plastic	$+0.087^{***}$	(0.028)	8.3	$+0.718$

Notes: $\Delta\beta^M$ is the industry-specific change in mature employment-growth responsiveness between the 2000s and the 2010s, estimated from the age-specific specification with three-digit within-industry productivity. "Contribution" is the employment-share-weighted contribution to the aggregate mature decline, in percentage points; industries are ordered by contribution. Standard errors clustered at the establishment level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

B.8 Relative comparative advantage and the remaining trade figures

Table 8 reports three trade-exposure measures in the main text and holds out a fourth, the change in Korea's revealed comparative advantage relative to China, $\Delta(\text{RCA}_{\text{KOR}}/\text{RCA}_{\text{CHN}})$.

Table B10: Correlation of relative comparative advantage with industry size

Size proxy	Correlation with $\Delta(\text{RCA}_{\text{KOR}}/\text{RCA}_{\text{CHN}})$
Establishment count (log)	−0.60
2001 shipment value (log)	−0.55
2001 absorption (log)	−0.66
Manufacturing employment share (log)	−0.61

Notes: Correlations across the eighteen harmonised manufacturing groups underlying Table 8. All four proxies are themselves correlated at 0.72–0.90 in logs, so this is one size fact checked four ways rather than four independent checks.

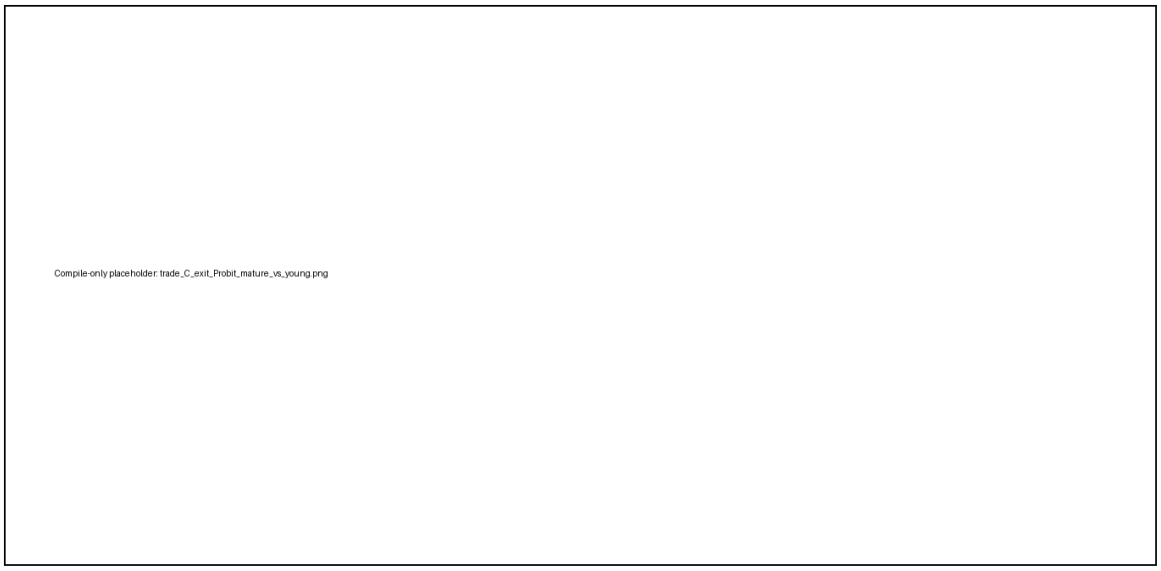


Figure B2: Exit Selectivity and Relative Comparative Advantage, by establishment age

Notes: A negative slope indicates that industries with rising relative comparative advantage against China saw *less* weakening of exit selectivity. This is the measure held out of Table 8 on mean-independence grounds; it is shown for transparency and carries no weight in the argument.

Unlike the other three, this measure is correlated with industry size across four independent proxies—establishment count, 2001 shipment value, 2001 absorption, and manufacturing employment share—at $|r| = 0.55$–0.66 (Table B10). A pooled cross-industry slope is interpretable as an average of industry-specific effects only under a mean-independence condition that this measure fails, so we report it here rather than in Table 8: the slope on the scale-free exit margin is −0.186 (mature) and −0.136 (young), both individually more precisely estimated than the three main-text measures, which is itself a symptom of the same problem rather than a virtue.

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